

Optimizing Agricultural Resource Allocation through Reinforcement Learning for Yield Improvement: A Cost-Driven Approach to Crop Efficiency Enhancement in Pakistan



By

Haider Masood

(Registration No: 401636)

Department of Computing

School of Electrical Engineering and Computer Science (SEECS)

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

March 2026

Optimizing Agricultural Resource Allocation through Reinforcement Learning for Yield Improvement: A Cost-Driven Approach to Crop Efficiency Enhancement in Pakistan



By

Haider Masood
(Registration No: 401636)

A thesis submitted to the National University of Sciences and Technology (NUST),
Islamabad, in partial fulfillment of the requirements for the degree of

Master of Science in Artificial Intelligence

Supervisor: Dr. Zuhair Zafar
Co-Supervisor:

School of Electrical Engineering and Computer Science (SEECS)
National University of Sciences and Technology (NUST)
Islamabad, Pakistan

March 2026

Thesis Acceptance Certificate

Certified that the final copy of the MS thesis written by Haider Masood (Registration No. 401636), of School of Electrical Engineering and Computer Science (SEECs), has been vetted in all respects as per prevailing NUST statutes and graduate policy requirements. The document is considered complete for evaluation, and the amendments advised by the Graduate Evaluation Committee may be incorporated in the final signed copy submitted to the university.

Signature: _____

Name of Supervisor: Dr. Zuhair Zafar

Date: _____

Signature (HoD): _____

Date: _____

Signature (Dean/Principal): _____

Date: _____

Form TH-4

National University of Sciences and Technology

MS Thesis Work

We hereby recommend that the thesis prepared under our supervision by Haider Masood, titled “Optimizing Agricultural Resource Allocation through Reinforcement Learning for Yield Improvement: A Cost-Driven Approach to Crop Efficiency Enhancement in Pakistan,” be accepted in partial fulfillment of the requirements for the award of the degree of Master of Science in Artificial Intelligence.

Examination Committee Members

1. Name: _____ Signature: _____

2. Name: _____ Signature: _____

3. Name: _____ Signature: _____

4. Name: _____ Signature: _____

Supervisor Dr. Zuhair Zafar Signature: _____

Declaration

I, Haider Masood, hereby declare that this thesis is my own work, prepared under the guidance of my supervisor, and has not been submitted in whole or in part for any other degree or diploma. All information drawn from published and unpublished sources has been acknowledged through proper citation. The technical claims in this draft are restricted to implemented repository features and completed research artifacts available at the time of writing.

Signature: _____

Date: _____

Acknowledgements

I am grateful to my supervisor, Dr. Zuhair Zafar, for the academic guidance that shaped this work from proposal to implementation. I also acknowledge the faculty and evaluators whose feedback during the synopsis and proposal-defense stages helped sharpen the technical scope of the thesis. This work benefited from prior open research on CyclesGym and from official data sources maintained by Pakistan government institutions, FAO, NASA POWER, and ISRIC SoilGrids. Finally, I acknowledge the support of my family and peers for their patience during the long experiment-engineering and writing cycles required to transform the original base repository into a thesis-ready research stack.

Abstract

This thesis develops and documents a Pakistan-adapted reinforcement learning workflow for long-horizon crop-management experimentation on top of the CyclesGym environment. The work begins from the original CyclesGym base repository and extends it in five implementation phases: Pakistan crop-calendar alignment, Pakistan price localization, multi-nutrient fertilization support, hierarchical integration of yearly crop planning with within-season fertilization, and standardized reporting for thesis-grade evaluation. The implemented stack uses a Pakistan weather file, Pakistan soil profile, and yearly crop and nutrient price series reconstructed from official sources, while keeping the active experimental crop setup on the current maize-soy configuration that is already wired into the repository.

The thesis adopts a hybrid scope. The synopsis is used as the motivational and planning baseline, but empirical claims are restricted to implemented controls and verified artifacts only. Consequently, the study focuses on crop planning, NPK-aware fertilization, and hierarchical reinforcement learning, while irrigation control, rice-specific localization, and field validation are retained as future work. The latest broad experiment matrix for the Pakistan NPK setup has already been encoded in the repository and standardized through a unified runner with per-run summary JSON outputs, but the full fresh campaign remains pending at the time of this draft build. For that reason, the thesis reports the complete system architecture, methodology, reproducibility design, and historical benchmark context from older audited runs, while explicitly separating those benchmark results from the new NPK campaign that is intended to supply the final comparative chapter.

The main contribution of this work is therefore twofold: first, it transforms the original CyclesGym codebase into a localized, NPK-capable, hierarchically extensible research platform for Pakistan-relevant decision support; second, it defines a defensible thesis-writing and experiment-reporting workflow in which every major claim is anchored either in completed code, completed artifacts, or clearly bounded future work. This structure prevents over-claiming while preserving the scientific value of the implemented engineering advances.

List of Abbreviations

A2C	Advantage Actor-Critic
API	Application Programming Interface
CGM	Crop Growth Model
CMDP	Constrained Markov Decision Process
CYCLES	Crop and environmental systems simulator wrapped by the repo
FAO	Food and Agriculture Organization of the United Nations
FAOSTAT	FAO statistical data platform
GDP	Gross Domestic Product
MDP	Markov Decision Process
NASA POWER	Prediction of Worldwide Energy Resources data service
NFD C	National Fertilizer Development Centre
NPK	Nitrogen, phosphorus, and potassium nutrient channels
PBS	Pakistan Bureau of Statistics
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
SB3	Stable-Baselines3
SEecs	School of Electrical Engineering and Computer Science
W&B	Weights and Biases

Contents

1	Introduction, Motivation, and Thesis Scope	1
1.1	Why a Simulation-First RL Thesis is Defensible	1
1.2	Pakistan Context and Input-Cost Motivation	2
1.3	Operational Decomposition of the Thesis Problem	4
1.4	Problem Statement	5
1.5	Research Questions	5
1.6	Evidence Hierarchy Used in This Document	6
1.7	Contributions of the Thesis	6
1.8	Scope Boundaries	7
1.9	Organization of the Thesis	8
2	Background and Literature Review	9
2.1	Crop-Growth Models and Long-Horizon Management	9
2.2	RL Foundations Relevant to the Thesis	9
2.2.1	Partial Observability and Delayed Credit	10
2.3	Cost-Aware and Constraint-Aware RL in Agriculture	11
2.3.1	Why Cost Awareness Is Central in the Pakistan Setting	11
2.4	Agricultural RL Environments Relevant to This Thesis	11
2.4.1	Why CyclesGym Was a Better Base Than Rebuilding from Scratch	13
2.5	Data Localization as a Research Requirement	13
2.6	Why the Thesis Uses a Hybrid Scope	14
2.7	What the Literature Implies for Algorithm Choice	14
2.8	Synthesis of the Literature for the Present Work	15
3	From the Base Repository to the Pakistan-Adapted Thesis System	16
3.1	Starting Point: the Original Base Repository	16
3.2	What Changed in the Thesis Repository	16
3.3	Architecture of the Current Thesis Stack	17
3.4	Simulation Lifecycle and Execution Semantics	19
3.4.1	Class Responsibilities and Separation of Concerns	20
3.5	Five Completed Implementation Phases	20

3.5.1	Why the Change Logs Matter as Internal Evidence	21
3.6	Localization Layers Added by the Thesis	21
3.7	From Nitrogen-Centric Logic to NPK-Aware Logic	22
3.8	Hierarchical Integration as a Thesis-Level Extension	22
3.9	Why Engineering Traceability Matters for the Thesis	23
4	Methodology and Experimental Protocol	24
4.1	Problem Formulation	24
4.2	Fertilization Environment	24
4.3	Crop-Planning Environment	25
4.4	Hierarchical Formulation	25
4.5	Observation, Action, and Reporting Design	27
4.6	Algorithms	28
4.7	Latest Broad NPK Matrix	29
4.8	Hypotheses	31
4.9	Metrics	31
4.10	Claims Policy and Statistical Discipline	32
5	Implementation, Reproducibility, and Evaluation Setup	33
5.1	Software and Repository Stack	33
5.2	Data Inputs and Provenance	33
5.3	Training and Evaluation Pipeline	35
5.4	Reporting Pipeline	37
5.4.1	Smoke-Run Design	37
5.5	Historical Runtime Budget and Campaign Planning	38
5.6	Reproducibility Controls	38
5.7	Current Build-Time Status	39
5.8	Why This Chapter Matters for the Final Thesis	40
6	Current Results Status and Historical Benchmark Context	41
6.1	A Necessary Note on the State of the Main Results Chapter	41
6.2	Status of the Fresh Broad NPK Matrix	41
6.3	Historical Benchmark Context from the February 2026 Audit	42
6.4	Interpretation of Historical Trends	44
6.5	What the Fresh Campaign Must Still Deliver	44
6.6	Threats to Validity at Build Time	45
6.7	Conclusion of the Provisional Results Chapter	45

7	Discussion, Limitations, Synopsis Gap Closure, and Future Work	46
7.1	What the Thesis Has Actually Achieved	46
7.2	Why Scope Control is a Scientific Strength	46
7.3	Limitations	48
7.4	Implications for a Thesis Defense	49
7.5	Future Work	50
7.6	Final Reflection	51
A	Full Broad Latest-NPK Matrix Listing	52
B	Reproducibility and Build Notes	56
B.1	Thesis Build Commands	56
B.2	Smoke-Run Script	56
B.3	Why Smoke Runs Precede the Full Matrix	56
B.4	Why the Current Draft Keeps Chapter 6 Provisional	57
B.5	Administrative Metadata	57
B.6	Rebuild Philosophy	57
C	Source and Artifact Map	58
C.1	How to Read the Source Map	59
D	Verification Snapshot	60
D.1	Interpretation of the Verification Checks	60
D.2	Residual Verification Gaps	61
E	Data Provenance and Output Schema	62
E.1	Why Provenance and Schema Belong in the Thesis	63

List of Figures

1.1	Pakistan crop price series used by the thesis stack, generated from the repository’s official-source-derived yearly price file.	2
1.2	Pakistan nutrient price series used by the thesis stack. These series motivate a cost-aware, rather than yield-only, optimization objective.	3
1.3	Coverage of the Pakistan weather file wired into training and evaluation. The thesis uses this file as the local weather backbone for the implemented environments.	3
1.4	Monthly climatology implied by the Pakistan weather file used by the thesis stack. The figure clarifies the seasonal structure against which weekly and yearly decisions are learned.	4
2.1	RL interaction loop as instantiated by the thesis stack. The loop is generic in theory but concrete in implementation: actions are mapped into simulator operations and observations are reconstructed from generated output files.	10
2.2	Pakistan crop-calendar windows currently encoded in the repository. This is an example of how localized agronomic priors enter the engineering stack without being overstated as full agronomic validation.	14
3.1	Capability evolution from the original base repository to the current thesis stack. The progression is defined by completed engineering phases, not by narrative aspiration alone.	17
3.2	Component-level architecture of the current thesis stack. The diagram highlights the interaction between policies, environments, simulator interfaces, and thesis reporting artifacts.	18
3.3	Sequence of one environment interaction cycle, from reset and input-file creation to simulator execution, parsing, and reward calculation.	20
4.1	Hierarchical decision flow implemented in the current repo. Annual crop-planning actions configure the season, while weekly fertilization actions operate within that season.	26

4.2	Decision timescales represented by the active thesis environments. The figure shows why yearly planning and weekly fertilization are modeled separately rather than forced into one homogeneous decision layer.	26
4.3	Reward decomposition used by the thesis environments. Harvest revenue and nutrient costs are computed by separate rewarder components and then combined.	28
4.4	Composition of the latest broad NPK matrix by domain and algorithm family.	30
4.5	Budget distribution of the latest broad NPK matrix. Fertilization budgets are expressed in total-years, while crop-planning jobs retain the current default timestep configuration.	30
5.1	Data provenance flow from official sources to local thesis artifacts. The figure clarifies how source documents become executable weather, calendar, and pricing inputs.	34
5.2	Training and evaluation pipeline used by the thesis experiments. The same pipeline is also responsible for the summary JSON outputs consumed by the writing workflow.	36
5.3	Reporting and thesis-asset generation workflow. This pipeline makes the final document reproducible from experiment outputs rather than manual transcription.	37
5.4	Historical runtime profile derived from earlier audited runs. This figure is used for campaign planning rather than for the main results claim.	38
5.5	Current status of the latest broad NPK matrix at the time of this build. . .	40
6.1	Historical fertilization benchmark context from earlier audited runs. These scores are retained only as context and should not be confused with the pending fresh broad NPK campaign.	42
6.2	Historical crop-planning benchmark context from earlier audited runs. As above, the figure is contextual rather than final evidence for the latest stack.	42
6.3	Failure signatures observed during the earlier audited experiment window. These signatures are useful for campaign planning and risk management during the fresh run phase.	43
7.1	Roadmap implied by the current thesis limitations. The roadmap is staged so that empirical completion, control expansion, and external validation occur in a defensible order.	51

List of Tables

1.1	Synopsis closure matrix used to keep thesis claims aligned with the implemented repo state.	7
2.1	Positioning of the thesis stack against closely related agricultural RL environments.	12
3.1	File-level responsibilities of the core thesis components.	19
3.2	Implementation phases already completed in the current repo before the fresh broad experiment campaign.	21
4.1	Active environment designs used in the thesis workflow.	27
4.2	Algorithm roles in the latest thesis experiment matrix.	29
4.3	Full broad latest-NPK matrix encoded in <code>run_experiments_7_3_2026.py</code>	30
5.1	Core artifacts that make the thesis stack reproducible.	34
5.2	Data provenance summary for the localized thesis stack.	35
5.3	Machine-readable outputs that connect experiments to thesis tables and figures.	36
5.4	Historical runtime statistics used to estimate the wall-clock budget of the fresh broad campaign.	38
5.5	Verification commands executed during the thesis workspace implementation on 8 March 2026.	39
5.6	Current status of the latest broad NPK matrix at the time of thesis build.	40
6.1	Current status of the latest broad NPK matrix at the time of thesis build.	41
6.2	Historical fertilization benchmarks from audited February 2026 runs.	43
6.3	Historical crop-planning benchmarks from audited February 2026 runs.	43
6.4	Historical failure signatures observed in the audited February 2026 runs.	44
7.1	Synopsis closure matrix used to keep thesis claims aligned with the implemented repo state.	47
7.2	Limitations matrix used to distinguish current evidence boundaries from future work.	49

A.1	Full broad latest-NPK matrix listing generated from the unified runner summary CSV.	52
C.1	File-level responsibilities of the core thesis components.	59
D.1	Verification commands executed during the thesis workspace implementation on 8 March 2026.	60
E.1	Data provenance summary for the localized thesis stack.	62
E.2	Machine-readable outputs that connect experiments to thesis tables and figures.	63

Chapter 1

Introduction, Motivation, and Thesis Scope

Pakistan’s agricultural sector faces a structurally difficult decision problem: managers must choose crops and input levels under uncertain weather, rising fertilizer prices, and delayed feedback from soil and yield outcomes. In practice, many farm-management decisions remain heuristic, calendar-driven, or based on short-horizon intuition rather than explicit long-term optimization. This creates a strong motivation for simulation-first decision support, especially where field experimentation is costly and risky. Official policy documents continue to place agriculture at the center of the national economy, while also emphasizing climate exposure, productivity pressure, and input-cost sensitivity [5]. In parallel, agricultural reinforcement learning (RL) has matured from a niche idea into a credible research direction for sequential decision-making in crop management [2].

The present thesis is motivated by a gap between that global research direction and the concrete state of local, Pakistan-grounded tooling. Existing benchmark environments such as CropGym and gym-DSSAT have advanced the field, but the repository used in this thesis was originally based on CyclesGym because CyclesGym is more naturally aligned with multi-year crop-management questions, soil carry-over effects, and rotation-aware reasoning [8, 3, 14]. However, the original base repository was not ready to serve as a Pakistan-focused thesis artifact without substantial additional engineering. It needed localized price series, clearer scope boundaries, reporting outputs suitable for academic defense, and explicit traceability from code changes to thesis claims.

1.1 Why a Simulation-First RL Thesis is Defensible

Simulation-first RL is defensible in agriculture for two reasons. First, agricultural decision quality is usually evaluated only after a season or multiple seasons, which makes exhaustive field experimentation impractical. Second, crop growth models create a repeatable environment in which policy learning can be studied before any pilot deployment. This does not eliminate the need for field validation, but it changes the stage at which expensive field testing is required. Instead of using the field to discover whether a policy family is viable at all, simulation can be used to narrow the search space to a smaller number of

plausible policies [14, 2].

The thesis therefore adopts a research posture that is realistic rather than promotional. It does not claim that simulation replaces agronomy or field trials. Instead, it treats the simulator as a controlled environment for learning decision policies, benchmarking algorithmic configurations, and stress-testing reward designs. This distinction is especially important because the repository is not a generic agricultural AI platform. It is a specific RL-enabled wrapper around the CYCLES simulator, and the defensibility of the thesis depends on keeping every claim anchored to what that stack actually does.

1.2 Pakistan Context and Input-Cost Motivation

A key thesis argument is that economic localization matters. It is not enough to optimize yield using generic or imported assumptions if the deployed environment is intended to represent Pakistan-relevant decisions. The repository already contains a Pakistan weather file, Pakistan soil profile, and a generated yearly price series derived from official data channels including FAOSTAT and NFDC-linked fertilizer prices. Figures 1.1, 1.2, and 1.3 summarize the localized signals that now shape the active thesis stack.

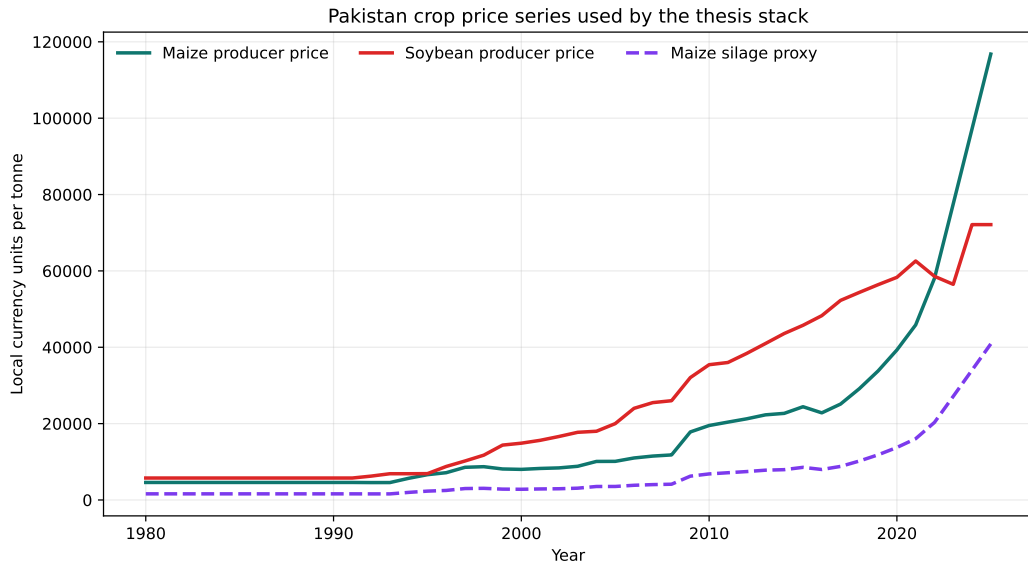


Figure 1.1: Pakistan crop price series used by the thesis stack, generated from the repository’s official-source-derived yearly price file.

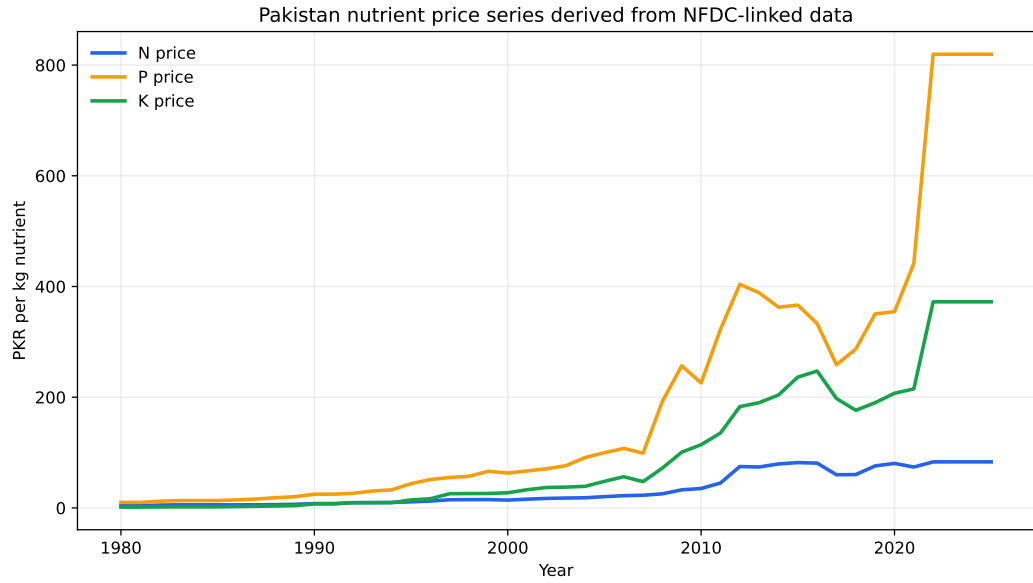


Figure 1.2: Pakistan nutrient price series used by the thesis stack. These series motivate a cost-aware, rather than yield-only, optimization objective.

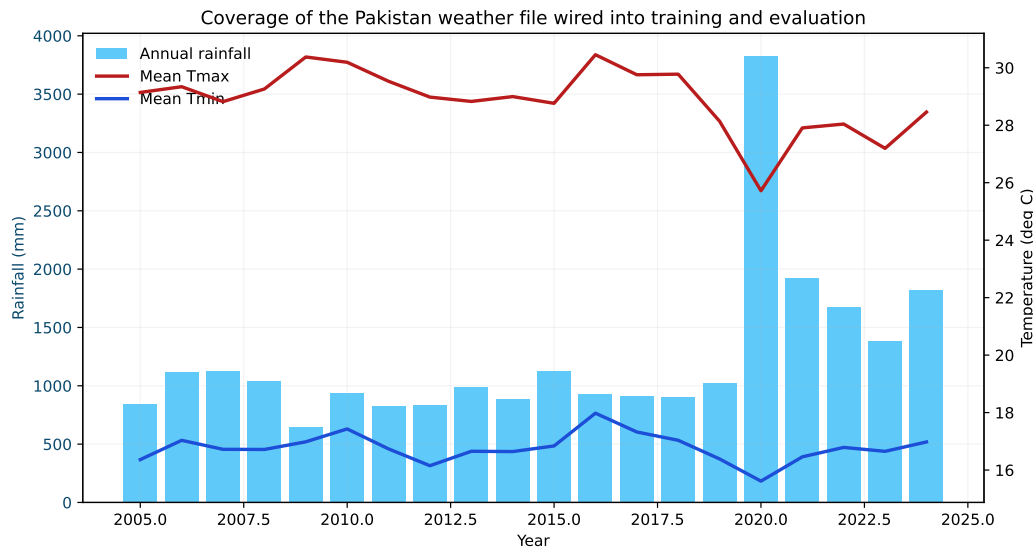


Figure 1.3: Coverage of the Pakistan weather file wired into training and evaluation. The thesis uses this file as the local weather backbone for the implemented environments.

These figures do not by themselves prove agronomic optimality. Their role is narrower and more important: they show that the repository now operates on localized inputs instead of remaining tied to generic defaults. That localization matters because the thesis is explicitly framed around Pakistan, and local economic conditions change both the meaning of reward design and the interpretation of policy quality.

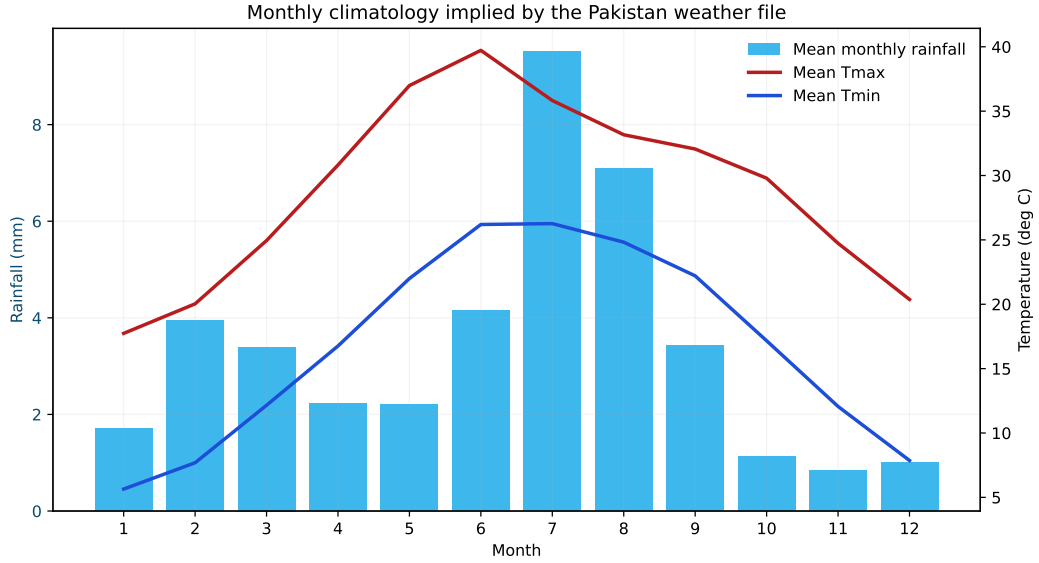


Figure 1.4: Monthly climatology implied by the Pakistan weather file used by the thesis stack. The figure clarifies the seasonal structure against which weekly and yearly decisions are learned.

The climatology perspective is important because the thesis is not optimizing in a stationary tabular world. The simulated environment reflects a strong seasonal structure, with rainfall concentration and temperature shifts that determine planting windows, crop development, and the timing sensitivity of nutrient application. A weekly RL policy that ignores this seasonal rhythm would not be scientifically meaningful even if it achieved a numerically high reward inside a narrow benchmark configuration.

1.3 Operational Decomposition of the Thesis Problem

The thesis decomposes the agricultural decision problem into three layers. The first layer is *data realism*: weather, soil, and price inputs must be localized enough that the reward signal carries Pakistan-relevant meaning. The second layer is *control design*: the environment must expose actions that correspond to recognizable management decisions such as crop choice, planting timing, and nutrient application. The third layer is *evidence discipline*: the training pipeline must produce outputs that allow the thesis to reconstruct what happened during learning and evaluation rather than only the final return.

This decomposition is one reason the current thesis is organized as a systems thesis as much as an algorithm thesis. A weak data layer would make cost-aware claims fragile. A weak control layer would make the learned policy difficult to interpret agronomically. A weak evidence layer would make the results chapter vulnerable to committee criticism because the thesis could not explain why a particular score occurred. The repository changes described in later chapters should therefore be read as targeted interventions on these three layers rather than as miscellaneous software improvements.

1.4 Problem Statement

The core problem addressed in this thesis is the following: how can a CYCLES-based RL environment be transformed into a Pakistan-adapted, cost-aware experimental platform that supports credible long-horizon crop-management research without overstating what has and has not been implemented? The question is not just whether an RL agent can achieve a high score. The harder question is whether the surrounding system, including data sources, action design, reward formulation, reporting outputs, and experiment runner, is rigorous enough to support a thesis that will be examined on both engineering and scientific grounds.

This problem has several dimensions. At the algorithmic level, the repository must support policies for sequential management decisions such as fertilization and crop planning. At the data level, it must replace legacy assumptions with Pakistan-relevant weather, soil, and price signals. At the reporting level, it must expose not only aggregate returns but also the intermediate decisions that matter to a thesis committee: which crop was selected, how much nutrient was applied, what the cost decomposition was, and whether a planting decision respected seasonal windows. At the academic-writing level, it must distinguish between implemented capability, planned experimentation, and future work.

1.5 Research Questions

The thesis is structured around four operational research questions.

1. How can the original CyclesGym base repository be localized for Pakistan-relevant crop-management experimentation while preserving backward compatibility where possible?
2. What architectural and reporting extensions are required to move from a largely nitrogen-centric workflow to a broader NPK-aware and hierarchical thesis stack?
3. How should the empirical chapter be structured when the latest broad NPK matrix is fully defined and ready to run, but not yet completed at the time of the current draft build?
4. Which thesis claims can be defended today as implemented contributions, and which must be retained as partial work or future extensions?

These questions are intentionally phrased to keep the thesis aligned with the actual repository state. The work is not presented as a completed farmer-advisory deployment, nor as a finalized field-validated agronomic product. It is presented as a thesis-grade research stack that has been materially extended and prepared for a broad experiment campaign.

1.6 Evidence Hierarchy Used in This Document

Not all evidence used in the thesis has the same epistemic role. Primary external literature establishes why simulation-based RL, crop-growth models, and cost-aware formulations are legitimate research directions. Official Pakistan and NUST sources establish the contextual and institutional grounding of the work. Repository artifacts establish what has actually been implemented. Experiment outputs, once the full campaign is run, will establish comparative performance within the defined scope.

This hierarchy matters because it prevents category mistakes. A proposal slide cannot substitute for a code implementation. A unit test cannot substitute for a completed experiment matrix. A historical benchmark from an older stack cannot substitute for the final latest-NPK evidence. The writing strategy of the thesis deliberately follows this hierarchy so that each claim is backed by the right kind of source.

1.7 Contributions of the Thesis

The first contribution is technical localization. The repository now includes a Pakistan-oriented weather and soil setup, crop-calendar enforcement hooks, and yearly price series that shift the reward design away from imported default assumptions. The second contribution is representational expansion. The fertilization pipeline is no longer restricted to a simple nitrogen-only framing; the codebase now supports NPK-aware logic and price decomposition. The third contribution is hierarchical integration. The system now supports a yearly crop-planning layer coupled with within-season fertilization decisions, which better reflects the multiscale nature of real farm management. The fourth contribution is reproducibility and reporting. Standardized summary JSON files, reporting callbacks, and a unified experiment runner make the results chapter reproducible rather than anecdotal.

A fifth contribution is methodological honesty. This thesis does not collapse planning documents, code changes, and completed results into a single unqualified narrative. Instead, it explicitly separates the synopsis promises from the implemented repository scope. That distinction is essential because the synopsis included ambitions such as irrigation control and rice-wheat localization that are not yet implemented in the active repo. The thesis therefore includes a synopsis closure matrix so that each promise is classified as implemented, partial, or future work.

Taken together, these contributions mean that the thesis should be evaluated on two axes. The first is whether the engineering stack is meaningfully better aligned with Pakistan-grounded, cost-aware agricultural RL than the inherited base repository. The second is whether the document preserves a disciplined boundary between implemented capability and deferred research ambition. The current draft argues that the answer is yes on both axes, even though the final broad empirical campaign is still pending.

Table 1.1: Synopsis closure matrix used to keep thesis claims aligned with the implemented repo state.

Synopsis item	Status	Evidence in the current repo	Thesis treatment
Pakistan-adapted weather and soil inputs	Implemented	Pakistan weather and soil files are wired into train and evaluation entypoints.	Present as a completed contribution.
Cost-aware reward shaping	Implemented	Reward stack and Pakistan price profiles are implemented and tested.	Present as a completed contribution.
Multi-nutrient fertilizer modeling	Implemented	NPK action mode, price series, and reporting are implemented.	Present as a completed contribution.
Hierarchical RL across planning and fertilization	Implemented	Dedicated hierarchical environment and reporting call-back exist.	Present as a completed contribution with explicit scope bounds.
Irrigation as a learned action	Not implemented	No active training flow exposes irrigation as a learned control variable.	Present as future work only.
Rice-specific cultivar localization	Not implemented	Current crop-planning stack uses the working maize-soy configuration.	Present as future work only.
Rice-wheat experimental rotation	Not implemented	Crop-calendar support exists, but the main experiment setup remains maize-soy.	Present as a deferred extension, not a result claim.
Farmer-facing advisory rule export	Partial	CSV-style reporting exists, but no polished advisory layer is implemented.	Present as partial engineering output.
Formal price- and climate-shock campaign	Partial	The matrix runner is ready, but the broad fresh NPK campaign is still pending at build time.	Present as a planned experiment campaign.

1.8 Scope Boundaries

The most important boundary is that the thesis methods and claims reflect the current working implementation, not the broadest aspiration in the synopsis. The implemented empirical scope covers Pakistan-adapted weather and soil integration, crop planning, NPK-aware fertilization, hierarchical RL scaffolding, and the latest unified experiment matrix. The active crop-planning and hierarchical setup remains tied to the current maize-soy configuration already wired into the repository. By contrast, irrigation as a learned action,

rice-specific localization, rice-wheat experiments, and formal field validation remain out of scope for the implemented results.

This bounded scope is a strength rather than a weakness. It prevents the thesis from claiming more than the repository can support, and it creates a clear roadmap for post-thesis work. In a field where engineering complexity often obscures what is truly implemented, scope clarity is part of scientific rigor.

The maize-soy decision should be read in this light. It is not presented as the unique or final agronomic target for Pakistan. It is presented as the currently stable experimental configuration for which the code, tests, runner defaults, and reporting pipeline are all aligned. In other words, the thesis prioritizes internal coherence over premature expansion to more ambitious but less stable crop configurations.

1.9 Organization of the Thesis

Chapter 2 establishes the literature and methodological background, with particular attention to crop-growth models, RL, constrained optimization, and related agricultural environments. Chapter 3 explains how the original base repository was transformed into the current Pakistan-adapted thesis stack. Chapter 4 formalizes the methodological and experimental design, including state, action, reward, matrix structure, and hypotheses. Chapter 5 documents the implementation, data provenance, reporting pipeline, and reproducibility controls. Chapter 6 reports the current execution status of the latest NPK matrix and contextualizes it against historical benchmark evidence from the earlier audited runs. Chapter 7 discusses limitations, closes the loop on synopsis promises, and outlines future work.

Chapter 2

Background and Literature Review

2.1 Crop-Growth Models and Long-Horizon Management

Agricultural decision-making is fundamentally sequential. A fertilizer decision made today affects not only near-term crop response but also subsequent nutrient availability, residual soil conditions, and risk exposure under weather uncertainty. This is one reason crop-growth models (CGMs) remain attractive in agricultural decision support. They provide a process-aware simulator in which weather, soil, crop physiology, and management interact over time. For RL researchers, CGMs are valuable because they make it possible to generate many simulated episodes without incurring the full cost of field experimentation [2, 14].

Yet not all CGM-based RL environments support the same research questions. Some are geared toward single-season nitrogen scheduling, others toward daily control, and others toward multi-year management with soil carry-over effects. This distinction matters for the present thesis because the long-horizon and rotation-aware behavior of CyclesGym is one of the reasons it was chosen as the thesis base.

2.2 RL Foundations Relevant to the Thesis

In the standard RL formulation, an agent observes a state or observation, selects an action, and receives a reward signal that reflects the quality of that action in context [13]. For agricultural management, this formulation is compelling because actions must often be chosen before the final effect of those actions is fully known. Fertilizer applied early in a season may increase yield, waste cost, or contribute to environmental losses depending on subsequent weather. Likewise, crop choice affects not only one season’s return but also the future soil and nutrient environment available to later crops.

Formally, the thesis uses the language of Markov decision processes (MDPs), and in some places the broader vocabulary of constrained or partially observed settings is more appropriate. The repository’s environments expose observations rather than complete physical states, and some important agronomic objectives such as nutrient losses or

compliance with seasonal windows are more naturally modeled as constraints or side metrics than as a single scalar reward. This is why constrained and safe RL literature is relevant even though the current training entrypts mainly optimize scalar reward functions rather than a full CMDP objective [6, 15].

RL interaction loop instantiated by the thesis stack

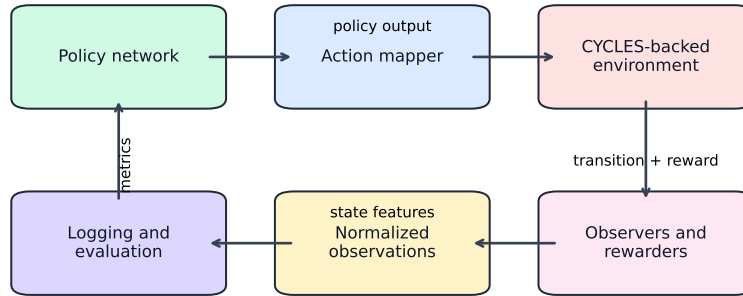


Figure 2.1: RL interaction loop as instantiated by the thesis stack. The loop is generic in theory but concrete in implementation: actions are mapped into simulator operations and observations are reconstructed from generated output files.

Figure 2.1 emphasizes a point that is easy to miss in high-level RL descriptions. In this repository, the transition dynamics are not an abstract simulator hidden behind a clean API. They are realized through file creation, CYCLES execution, output parsing, normalization, and callback-based logging. This makes agricultural RL both richer and harder than many benchmark-control tasks. The consequence for the thesis is that algorithm selection cannot be discussed independently of engineering reliability.

2.2.1 Partial Observability and Delayed Credit

Agricultural RL contains two difficulties that deserve explicit mention. The first is partial observability. The agent does not directly observe the full biophysical state of the soil-plant-atmosphere system. It receives a subset of features reconstructed from weather, crop, and soil output files. The second is delayed credit assignment. A fertilizer action taken early in a season may not be reflected in realized harvest revenue until much later, and even then its effect is entangled with weather and other management decisions. These properties help explain why stable policy-gradient methods remain attractive in the literature even when the action spaces are discrete.

2.3 Cost-Aware and Constraint-Aware RL in Agriculture

A common simplification in early agricultural RL work is to optimize productivity without fully localizing costs or explicitly tracking environmental side effects. That simplification may be useful for proof-of-concept work, but it becomes problematic for a thesis that claims cost-driven decision support. The present work therefore emphasizes cost-aware design in two ways. First, reward design is explicitly discussed as a revenue-minus-cost formulation rather than as a pure yield target. Second, the reporting layer preserves side metrics that are not collapsed into a single reward number, including nutrient-wise cost decomposition and season-window compliance.

This framing is consistent with broader constrained RL literature, which argues that practical decision systems often need to preserve auxiliary performance signals rather than bury them entirely inside one scalar target [6, 15]. In the agricultural domain, this need is even sharper because agronomic plausibility, environmental impact, and economic return are often coupled but not identical.

2.3.1 Why Cost Awareness Is Central in the Pakistan Setting

For the present thesis, cost awareness is not a generic design preference. It is part of the research claim implied by the title itself. A yield-only optimization would answer a different question from the one posed here. Once Pakistan-specific crop and nutrient prices are introduced, the reward becomes sensitive not only to agronomic productivity but also to the economic burden of nutrient application and the relative profitability of alternative crop decisions. This is especially important in a thesis framed around resource allocation rather than around maximum biological yield alone.

2.4 Agricultural RL Environments Relevant to This Thesis

The literature most directly relevant to this thesis includes three environment families. CropGym focuses on nitrogen management with crop-growth models and provides a clean benchmark for single-season scheduling [8]. gym-DSSAT turns DSSAT into an RL environment and demonstrates a different pathway for coupling agronomic simulation with agent learning [3]. CyclesGym, which is the foundation of this thesis, explicitly emphasizes long-term crop management strategies with support for multi-year dynamics and richer management structure [14].

The present thesis is not a direct reproduction of the original CyclesGym paper. Rather, it starts from the publicly available CyclesGym implementation and engineers a Pakistan-adapted, thesis-oriented stack around it. That engineering layer is what differentiates this work from a straightforward reimplement. It is also the reason that the literature

review must go beyond algorithm names and address environment fit, localization burden, and reporting needs.

Table 2.1: Positioning of the thesis stack against closely related agricultural RL environments.

Environment	Main scope	Strength for this thesis	Limitation for this thesis
CropGym [8]	Single-season nitrogen management on crop-growth simulations	Useful nitrogen-management benchmark with transparent reward trade-offs	Limited long-horizon rotation modeling and narrower carry-over effects
gym-DSSAT [3]	Daily control in DSSAT-based environments	High-fidelity crop-model integration and daily action interface	More limited rotation emphasis and a heavier integration footprint
CyclesGym [14]	Multi-year crop-management RL on the CYCLES simulator	Strong match to long-horizon crop rotation, soil C/N dynamics, and constraint-aware management	Requires more engineering to localize data and reporting for Pakistan-specific thesis work
This thesis stack	Pakistan-adapted CyclesGym with NPK and hierarchical RL extensions	Adds Pakistan price localization, crop-calendar alignment, hierarchical integration, and thesis reporting outputs	Latest full broad NPK campaign still needs completion before final empirical claims are frozen

Table 2.1 shows why CyclesGym was the most suitable base for this thesis. CropGym is strong for nitrogen scheduling, and gym-DSSAT is important as a DSSAT-based benchmark, but CyclesGym better matches the long-horizon, rotation-aware, and system-engineering needs of this project. The table also makes an important thesis point: even after choosing the right base environment, a large amount of local engineering is still required before the repository becomes a defensible thesis artifact.

2.4.1 Why CyclesGym Was a Better Base Than Rebuilding from Scratch

An alternative design path would have been to rebuild a Pakistan-specific environment from scratch on top of a crop model or a custom simulator wrapper. The thesis did not take that route because the research bottleneck was not the absence of any RL environment. The bottleneck was the absence of a localized, traceable, thesis-ready extension of an environment already suited to long-horizon crop management. Starting from CyclesGym preserved valuable prior structure such as multi-year simulation semantics, input-manager abstractions, and environment wrappers. The actual thesis contribution therefore lies in targeted adaptation rather than in unnecessary reinvention.

2.5 Data Localization as a Research Requirement

Localization in this thesis is not a cosmetic change. It is a research requirement. The moment the thesis title commits to Pakistan, the data sources and price assumptions become part of the scientific validity of the work. That is why the repository’s current implementation uses a Pakistan weather file, Pakistan soil profile, and yearly price series derived from FAOSTAT and NFDC-linked data. The weather file is the actual input used by training and evaluation flows, while the price series is loaded by the pricing utilities that underlie the current reward configuration.

The provenance of those inputs also matters. NASA POWER and ISRIC SoilGrids are cited because they represent standard, well-documented sources for weather and soil information in data-constrained settings [9, 7]. PBS crop calendars and Khyber Pakhtunkhwa maize guidance are cited because the current Pakistan crop-calendar utility in the repository depends on those official or semi-official references for sowing-window alignment [11, 12, 4]. The thesis does not treat these data products as perfect proxies for local agronomy; rather, it treats them as transparent and documentable sources that materially improve the realism of the training environment.

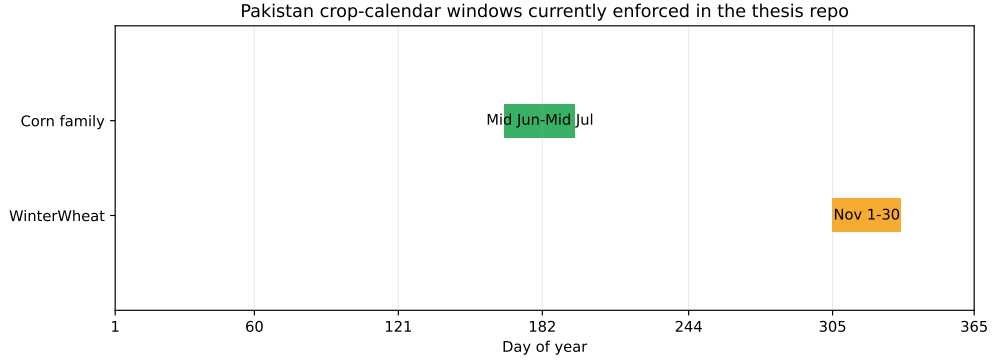


Figure 2.2: Pakistan crop-calendar windows currently encoded in the repository. This is an example of how localized agronomic priors enter the engineering stack without being overstated as full agronomic validation.

2.6 Why the Thesis Uses a Hybrid Scope

The synopsis used an intentionally broad framing that included irrigation, rice-wheat rotation, and broader Pakistan smallholder narratives. Such breadth is understandable at the proposal stage because the proposal must motivate a program of work rather than only the final implemented artifact. The thesis, however, must narrow its claims to what was actually built. This is why the literature review is organized around a hybrid scope. It preserves the broader background on agricultural RL and Pakistan-specific need, but it narrows the implementation and empirical interpretation to the controls that exist in the repository: crop planning, NPK-aware fertilization, and hierarchical integration.

This hybrid scope is academically preferable to two flawed alternatives. One flawed alternative would be to write only about the fully implemented repo and ignore the proposal context that motivated it. The other flawed alternative would be to write the thesis as if all proposal ambitions had been completed. The present thesis avoids both extremes by using the synopsis as the planning baseline and the repository as the claims boundary.

2.7 What the Literature Implies for Algorithm Choice

The literature does not dictate one universally best RL algorithm for agricultural management, but it does suggest a useful discipline for method selection. Methods should be chosen based on compatibility with the decision horizon, action structure, and available reporting, not only on popularity. In the current thesis stack, PPO is retained as the primary method because it historically behaved robustly in the repository and fits well with normalized, vectorized environments. A2C remains useful as a leaner actor-critic comparison. DQN is included only as a constrained ablation because the planning tasks require an action-space wrapper that changes the structure the value learner actually sees. This method selection logic is therefore an outcome of both literature and code reality.

2.8 Synthesis of the Literature for the Present Work

The literature suggests three conclusions that shape the rest of the thesis. First, RL is a credible approach to crop management when paired with a process-aware simulator and a carefully designed reward function [13, 2]. Second, the choice of simulator-wrapper environment materially changes what questions can be studied, and CyclesGym is the most appropriate base for the long-horizon thesis goals pursued here [14, 8, 3]. Third, localization and reporting are not optional add-ons in a thesis framed around Pakistan and cost-aware optimization; they are central research requirements.

These conclusions motivate the architecture and methodology presented in the next two chapters. Chapter 3 turns from external literature to the internal evolution of the repository, while Chapter 4 formalizes the resulting experimental design.

Chapter 3

From the Base Repository to the Pakistan-Adapted Thesis System

3.1 Starting Point: the Original Base Repository

The folder `Reference Documents` and `Folders/cyclesgym` base provides the original repository snapshot from which the present thesis work evolved. That base repository already exposed the central idea of CyclesGym: a reinforcement-learning wrapper around the CYCLES crop simulator. It contained environment definitions, training entrypoints, notebooks, and documentation, but it did not yet encode the full thesis-specific layer required for a Pakistan-grounded, defense-ready research artifact.

In the base form, the repository was useful as a simulator interface and experimentation framework. However, its structure was still closer to a research code release than to a thesis platform. Important thesis needs were either missing or underdeveloped: localized price profiles, explicit scope mapping against the proposal, broader nutrient support, standardized per-run summary outputs, and narrative documentation that could tie engineering changes to thesis claims.

3.2 What Changed in the Thesis Repository

The current repository is materially larger and more structured than the base version. New directories such as `final_docs`, `Changes`, `Experimentation` and `Results`, `wandb exports`, and the current thesis workspace itself reflect a transition from a generic research repo to a traceable thesis codebase. The new structure does not merely add notes. It adds engineering layers that are directly relevant to the thesis: localized pricing utilities, thesis reporting callbacks, a hierarchical environment, and a broad matrix runner for the latest Pakistan NPK design.

Figure 3.1 summarizes this transition at the capability level.

Capability evolution from the original base repo to the thesis stack

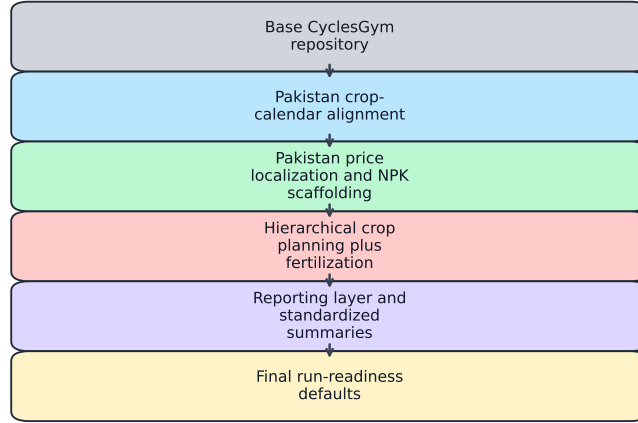


Figure 3.1: Capability evolution from the original base repository to the current thesis stack. The progression is defined by completed engineering phases, not by narrative aspiration alone.

The most important lesson from this evolution is that thesis readiness is not equivalent to simply running more experiments. It also requires building the surrounding apparatus that makes those experiments interpretable, reproducible, and bounded in scope.

3.3 Architecture of the Current Thesis Stack

At a high level, the current system has four layers. The top layer is the policy-learning layer, implemented through Stable-Baselines3. The second layer is the environment layer, where Corn, crop-planning environments, and the hierarchical environment map actions into simulator-relevant operations. The third layer is the simulation I/O layer, which writes control, operation, weather, and related files and invokes the CYCLES executable. The fourth layer is the parsing and reporting layer, which reconstructs observations, computes rewards, and writes thesis-grade outputs.

Component-level architecture of the thesis system

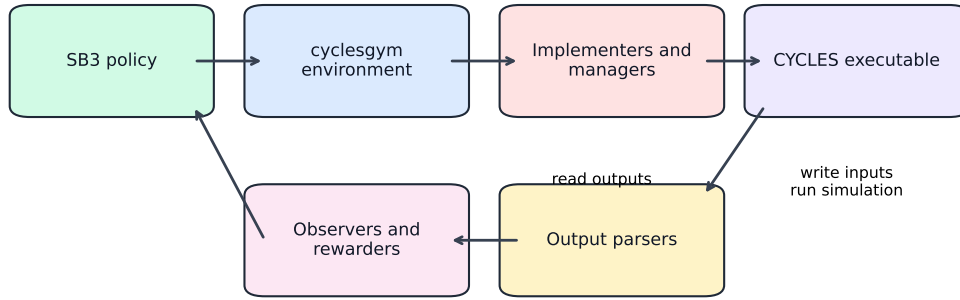


Figure 3.2: Component-level architecture of the current thesis stack. The diagram highlights the interaction between policies, environments, simulator interfaces, and thesis reporting artifacts.

The architecture matters because the thesis is not just about model performance. It is also about whether the stack is structured in a way that permits clear causal interpretation of results. If the simulator run, output parsing, or reporting layer is opaque, then even strong returns become difficult to defend. The current architecture improves that situation by making the system more modular and by attaching thesis-focused reporting outputs to the same environment steps used for training.

Table 3.1: File-level responsibilities of the core thesis components.

Component	Main file	Responsibility	Why it matters to the thesis
Weekly fertilization environment	<code>cyclesgym/envs/corn.py</code>	Maps weekly agent actions into fertilizer operations and simulator runs.	Main benchmark for Pakistan-aligned fertilization experiments.
Flat crop-planning environment	<code>cyclesgym/envs/crop_planning.py</code>	Encodes annual crop and planting choices with soil-nitrogen observations.	Strategic planning layer for the maize-soy thesis setup.
Hierarchical environment	<code>cyclesgym/envs/hierarchical.py</code>	Couples yearly crop planning with weekly NPK fertilization and rich info dictionaries.	Core thesis-specific system extension.
Pricing localization	<code>cyclesgym/utils/pricing_utils.py</code>	Loads Pakistan yearly crop and nutrient price series with legacy fallbacks.	Converts narrative cost-awareness into executable reward logic.
Reporting callback	<code>cyclesgym/utils/thesis_reporting.py</code>	Writes weekly nutrient logs, yearly crop decisions, compliance files, and summary JSON.	Makes Chapter 6 traceable to generated artifacts.
Experiment orchestrator	<code>run_experiments_7_3_2026.py</code>	Builds the 113-job matrix and standardizes per-run summary outputs.	Defines the authoritative latest-NPK campaign.

Table 3.1 turns the architecture narrative into a file-level map. This is important for thesis defense because the committee can ask not only what the system does, but where the relevant logic lives. A traceable answer is much stronger than an abstract architectural description.

3.4 Simulation Lifecycle and Execution Semantics

The thesis repository still depends on the same fundamental simulation loop: the environment writes inputs, invokes CYCLES, reads outputs, and returns observations and rewards. But the lifecycle has been hardened and documented more carefully than in the base repo, partly because integration issues and cross-platform execution paths had to be stabilized before the broader NPK experiment campaign could be treated seriously.

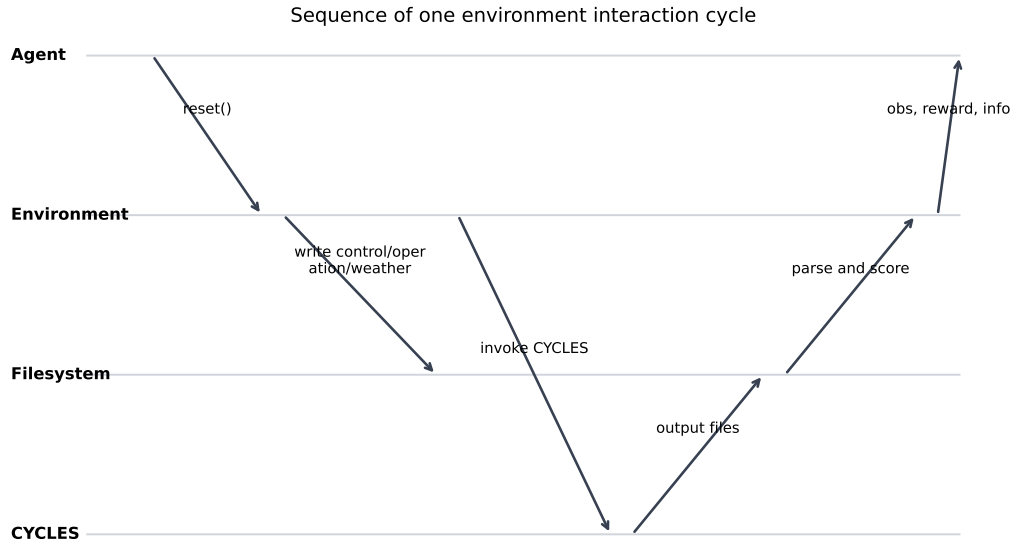


Figure 3.3: Sequence of one environment interaction cycle, from reset and input-file creation to simulator execution, parsing, and reward calculation.

This lifecycle is central to reproducibility. A thesis committee may reasonably ask whether a reported return is really linked to the intended combination of weather file, action implementer, and reward function. Because the system rewrites and re-parses simulator artifacts on every relevant step, the answer must be expressed as a concrete file-and-callback pipeline rather than as a vague software description.

3.4.1 Class Responsibilities and Separation of Concerns

The current repository separates concerns more clearly than the inherited base version. Observers convert simulator outputs into learning-state features. Implementers translate agent actions into operations-file edits. Rewarders calculate scalar economic signals from harvest and nutrient costs. Managers handle file-backed input and output data structures. This separation is not merely stylistic. It made it possible to add Pakistan price localization and thesis reporting without rewriting the entire environment stack.

3.5 Five Completed Implementation Phases

A large share of the thesis engineering effort is captured in the local change logs. These logs are not cited as external literature; instead, they function as internal traceability artifacts that justify why the thesis can describe particular capabilities as implemented. Table 3.2 condenses those records into a compact narrative.

Table 3.2: Implementation phases already completed in the current repo before the fresh broad experiment campaign.

Step	Theme	Outcome	Verification note
1	Pakistan crop-calendar alignment	Added conservative local sowing windows for maize and wheat families.	Targeted tests and backward-compatible defaults
2	Price localization and NPK scaffolding	Added Pakistan price profiles plus NPK-ready reward and action logic.	21 targeted tests passed
3	Hierarchical integration	Added yearly crop-planning plus weekly fertilization environment.	Full integration path stabilized across platforms
4	Reporting and data hardening	Added structured reporting CSVs, yearly series generation, and standardized summary JSON outputs.	Full test suite passed at this stage
5	Final NPK run readiness	Updated default training settings to Pakistan-aligned NPK mode and validated runner wiring.	59 tests passed and dry-run matrix validated

The table reveals an important pattern. The work progressed from realism to dimensionality to integration to reporting to run readiness. This is a more defensible trajectory than jumping directly into a large experiment matrix while the platform still lacks localized economics, multi-nutrient control, or structured reporting outputs.

3.5.1 Why the Change Logs Matter as Internal Evidence

The thesis uses the change logs in the `Changes/` directory as internal evidence of implementation chronology. They are not treated as external scholarly sources, but they serve an important methodological function: they preserve the order in which capabilities were added and reduce the risk of retroactively narrating the repository as if all features had existed from the beginning. This kind of engineering chronology is particularly valuable in research codebases where the final state can obscure the path by which stability was reached.

3.6 Localization Layers Added by the Thesis

The first localization layer is agronomic timing. The repository now includes a Pakistan crop-calendar utility that supports conservative sowing windows based on official or semi-official guidance. The second localization layer is economic. The price utilities no

longer rely only on legacy assumptions; they now load yearly Pakistan-relevant crop and nutrient price series. The third localization layer is narrative. The repo now explicitly documents what it does and does not optimize under the Pakistan framing.

These layers are conceptually separate. Crop calendars are about agronomic plausibility. Price series are about economic validity. Scope documentation is about scientific honesty. They should not be conflated, because each answers a different thesis-risk question.

Another localized design choice is the decision to keep the working experiments on the maize-soy configuration while still exposing crop-calendar hooks that can support broader future crop sets. This preserves forward compatibility without forcing the thesis to defend crop configurations that have not yet been stabilized in code and experiments.

3.7 From Nitrogen-Centric Logic to NPK-Aware Logic

One of the most important thesis-specific upgrades is the shift from a mostly nitrogen-centric pipeline to an NPK-aware pipeline. In the earlier state, many experiments and reward interpretations were naturally read as nitrogen optimization. In the current state, the fertilization environment, pricing utilities, and reporting layer can handle N, P, and K as separate channels. This is scientifically useful even before the full broad matrix is rerun, because it changes the modeling capacity of the system and the meaning of cost decomposition.

At the code level, this upgrade appears in multiple places at once. The `Corn` environment can now expose a `MultiDiscrete` NPK action space. The pricing utilities load nutrient-wise Pakistan series rather than a single scalar nitrogen cost. The reward layer computes cost channel by channel. The hierarchical reporting callback records `n_kg`, `p_kg`, `k_kg`, and their corresponding costs separately. The scientific importance of the NPK extension therefore lies not in one isolated code change, but in the way several modules now agree on the same nutrient vocabulary.

This upgrade does not automatically imply that the final empirical chapter can already claim broad NPK superiority. The full fresh campaign is still needed for that. But it does mean that the thesis can legitimately describe the current stack as NPK-capable rather than merely N-only.

3.8 Hierarchical Integration as a Thesis-Level Extension

The addition of the hierarchical environment is significant because it aligns the software representation more closely with how real farm management decisions are made. Some decisions, such as crop choice and planting windows, are yearly or seasonal. Others, such as nutrient application, are repeated within a season. A flat environment can model both

kinds of decisions, but a hierarchical structure makes the distinction explicit and easier to report.

That distinction is valuable in two ways. First, it supports clearer reasoning about what the policy is actually doing. Second, it creates a cleaner mapping between the technical stack and the story told in the thesis. A committee can understand annual planning and weekly fertilization as different but related decision layers. The code now reflects that intuition instead of leaving it implicit.

The hierarchical environment also improves thesis reporting. Its `info` dictionary now carries yearly crop-choice metadata, planting-window compliance, and nutrient cost breakdowns alongside the standard step outputs. This is important because it allows later analysis to separate “what crop was chosen” from “how much nutrient was applied” instead of compressing both into one final reward number.

3.9 Why Engineering Traceability Matters for the Thesis

A common weakness in experimental software theses is that the final written document refers to a polished conceptual system that the code never fully implemented. This thesis attempts to avoid that pattern by treating change logs, test outcomes, and generated tables as first-class evidence. The fact that the full test suite reached 59 passing tests before the fresh broad campaign is not a decorative software statistic. It is part of the argument that the platform is stable enough to support a defensible experiment phase.

The next chapter builds on this system view by formalizing the methodological choices encoded in the current stack, including state, action, reward, matrix structure, and reporting design.

Chapter 4

Methodology and Experimental Protocol

4.1 Problem Formulation

The thesis models crop-management decisions as sequential decision problems on top of the CYCLES simulator. The environments expose observations derived from simulator outputs and local configuration files, accept actions that modify management operations, and return scalar rewards together with richer reporting fields. In the simplest formulation, the agent seeks a policy π that maximizes expected discounted return

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t \right], \quad (4.1)$$

where r_t is a task-specific reward composed from crop revenue, input costs, and selected penalties or reporting-side constraints. This framing follows standard RL practice [13], but the thesis adds two important refinements. First, the reward is explicitly cost-aware rather than purely yield-based. Second, several agronomic and reporting signals are preserved outside the scalar reward so that the thesis can interpret behavior beyond aggregate return.

In practical terms, the problem is closer to a partially observed and weakly constrained MDP than to a perfectly observed one. The agent observes simulator-derived features rather than a complete agronomic state, and the thesis keeps some agronomic signals as side metrics instead of embedding all of them into the scalar reward. This choice is deliberate. It keeps the learning objective tractable while preserving the information required for later scientific interpretation.

4.2 Fertilization Environment

The fertilization environment operates at a weekly decision frequency. In its current thesis-ready form, the environment can operate in a single-nutrient nitrogen mode or in a multi-nutrient NPK mode, with the latter now serving as the localized default for the planned fresh campaign. The observation design combines weather information, crop-progress information, and cumulative nutrient-use information. The action design maps the policy

output into a discrete control over nutrient mass per hectare.

At a conceptual level, the weekly reward can be written as

$$r_t = \text{Revenue}_t - \text{Cost}_{N,t} - \text{Cost}_{P,t} - \text{Cost}_{K,t} - \text{Penalty}_t, \quad (4.2)$$

where the penalty term may include environmental or operational considerations depending on the environment configuration. The thesis does not claim that this reward fully captures all real-world farm costs. Instead, it claims that the reward is more localized and interpretable than the earlier legacy assumptions because it is tied to Pakistan-oriented yearly price series and can be decomposed by nutrient channel.

The code implements this structure through a compound rewarder. Harvest revenue is produced by `CropRewarder` when a harvest event occurs between successive dates. Nutrient cost is produced by `NProfitabilityRewarder` or `NPKProfitabilityRewarder`, depending on whether the environment is in scalar-N or full-NPK mode. The total step reward is the sum of these components. This means the thesis can describe the reward function without inventing a separate mathematical object that does not exist in code.

4.3 Crop-Planning Environment

The crop-planning environment operates at a slower timescale than fertilization. It selects yearly or seasonal crop-management actions, with the active working configuration focused on the current maize-soy rotation already wired into the repository. This crop choice is not presented as the final agronomic target for all Pakistan settings. Rather, it is presented as the working configuration for which the current code, tests, and latest matrix are aligned.

The crop-planning state abstracts longer-horizon information than the weekly fertilization state. It focuses on the information required to compare crop choices and planting-window decisions across seasons. Because the environment is already integrated with Pakistan crop-calendar logic, it can also surface whether a chosen action respects the configured seasonal windows.

The active flat training setup uses `CropPlanningFixedPlanting`, which reduces the action to crop choice plus a planting-week index. This design is important for experimental stability. It preserves the strategic nature of the planning problem while avoiding unnecessary expansion of the action space before the main broad matrix has been completed.

4.4 Hierarchical Formulation

The hierarchical environment combines yearly crop-planning actions with within-season fertilization actions. This yields a two-scale decision structure in which strategic and

tactical choices are represented explicitly rather than being flattened into one uniform action channel.

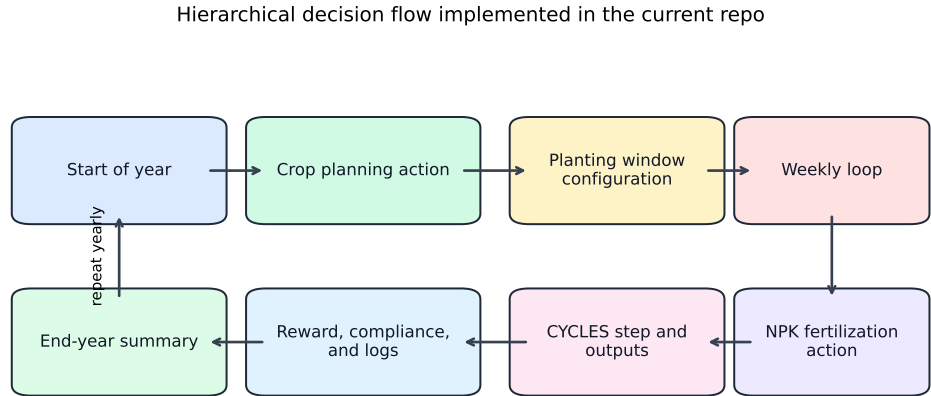


Figure 4.1: Hierarchical decision flow implemented in the current repo. Annual crop-planning actions configure the season, while weekly fertilization actions operate within that season.

The hierarchical structure is important for two reasons. First, it better matches the semantics of real management decisions. Second, it enables more interpretable reporting. A thesis reader can separately inspect yearly crop decisions, weekly nutrient actions, and their cost decomposition. That is an advantage over flatter designs in which the learning objective may remain valid but the resulting policy is harder to explain.

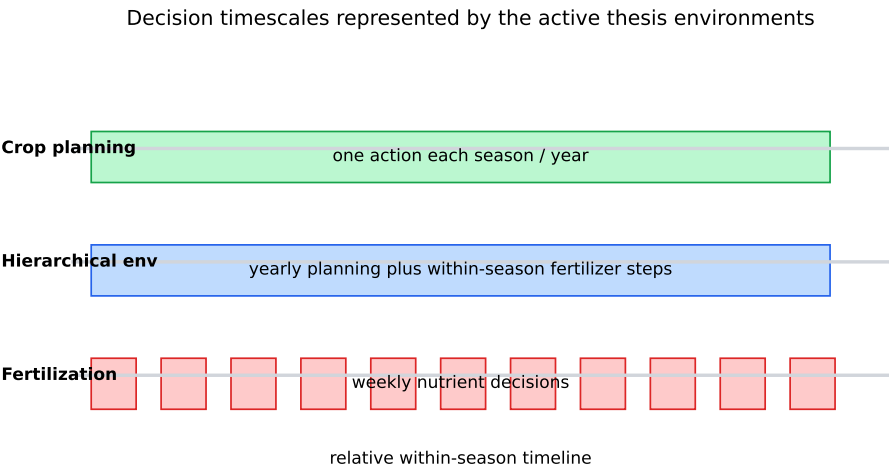


Figure 4.2: Decision timescales represented by the active thesis environments. The figure shows why yearly planning and weekly fertilization are modeled separately rather than forced into one homogeneous decision layer.

4.5 Observation, Action, and Reporting Design

The repository exposes different observation and action choices depending on the environment variant. Nevertheless, the thesis uses a consistent reporting vocabulary across environments:

- observations summarize relevant weather, crop, and nutrient information;
- actions encode either yearly planning choices or weekly fertilization choices;
- rewards quantify cost-aware performance;
- side metrics record nutrient-wise cost, crop decisions, and season-window compliance.

This design choice is methodologically useful because it avoids a common failure mode in RL theses: reducing all analysis to the final return metric even when the system already emits richer decision traces.

Table 4.1: Active environment designs used in the thesis workflow.

Environment	Decision interval	Observation design	Action design	Thesis use
Corn	7 days	27 features from weather, crop state, and cumulative nitrogen-to-date	Discrete N or 3-channel NPK fertilizer action	Main fertilization benchmark under Pakistan price and weather defaults.
CropPlanningFixedPlanting	1 year	11 soil-nitrogen features	Crop choice plus planting-week index	Main flat crop-planning baseline used by the working maize-soy setup.
HierarchicalCropPlanningFertilization	7-day loop with first-step yearly planning	14 features from soil nitrogen plus cumulative N and year index	7-channel action: crop, window start, window end, max SMC, N, P, K	Thesis-specific combined planner plus fertilization environment with report-rich outputs.

Table 4.1 summarizes the active environment designs in the same vocabulary used by the code. This is useful because the thesis needs to distinguish between the general environment family and the exact variants used in the current runner defaults.

Reward decomposition used by the thesis environments

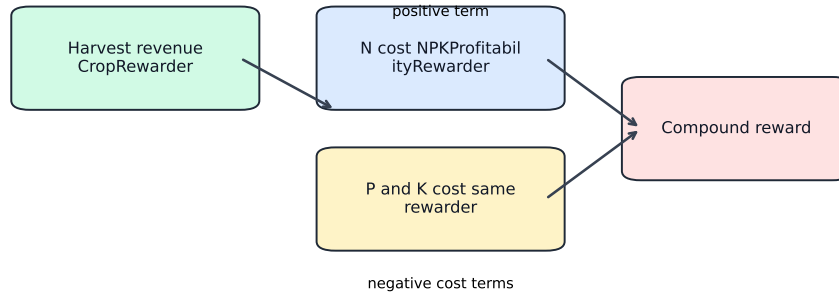


Figure 4.3: Reward decomposition used by the thesis environments. Harvest revenue and nutrient costs are computed by separate rewarder components and then combined.

The reporting design deserves separate emphasis. In the hierarchical environment, the `info` dictionary records whether the planner acted in the current step, what crop and planting window were selected, the amount of each nutrient applied, and the resulting cost breakdown. These fields are later streamed into CSV and JSON artifacts. The thesis therefore treats reporting fields as part of the experimental design rather than as a convenience afterthought.

4.6 Algorithms

The latest unified experiment matrix focuses primarily on PPO and A2C, with smaller DQN ablations and a baseline comparison path. PPO is treated as the main candidate because it has historically performed strongly in the repository’s earlier audited runs and is widely used for stable on-policy optimization in discrete or mixed decision settings [13]. A2C is retained as a simpler actor-critic baseline. DQN is included only as a narrower ablation because its compatibility with the crop-planning action structure requires additional wrapping and has historically shown more brittle behavior in the repo.

The thesis therefore uses algorithms not as an exhaustive zoo, but as a structured comparison between one primary method, one simpler actor-critic alternative, one narrower value-based ablation, and explicit open-loop or baseline-style references where available.

Table 4.2: Algorithm roles in the latest thesis experiment matrix.

Method	Primary use in this thesis	Rationale	Limitation
PPO	Main method across fertilization, crop planning, and hierarchical runs	Historically strongest method in repo audits and robust default for discrete policy optimization	On-policy training is comparatively expensive.
A2C	Secondary actor-critic baseline	Simpler policy-gradient baseline that shares much of the PPO stack	Usually less stable and lower performing than PPO in this repo.
DQN	Narrow ablation only	Tests whether a value-based learner remains competitive after action-space wrapping	Requires flattening MultiDiscrete actions and is more brittle in planning tasks.
Baseline policies	Reference comparison	Prevents RL scores from being interpreted without a non-learning point of comparison	Not intended to be a strong production policy family.

This limited algorithm set is a methodological strength. It avoids dispersing experimental effort across many methods while the more important variables of this thesis are localization, reporting, and cross-environment comparability.

4.7 Latest Broad NPK Matrix

The main experiment campaign for the thesis is the unified runner encoded in `run_experiments_7_3_2026.py`. This matrix is treated as authoritative because it standardizes budgets, algorithm families, hierarchical variants, baseline inclusion, and summary-JSON generation. The runner defines 113 jobs in total: 75 fertilization jobs and 38 crop-planning jobs.

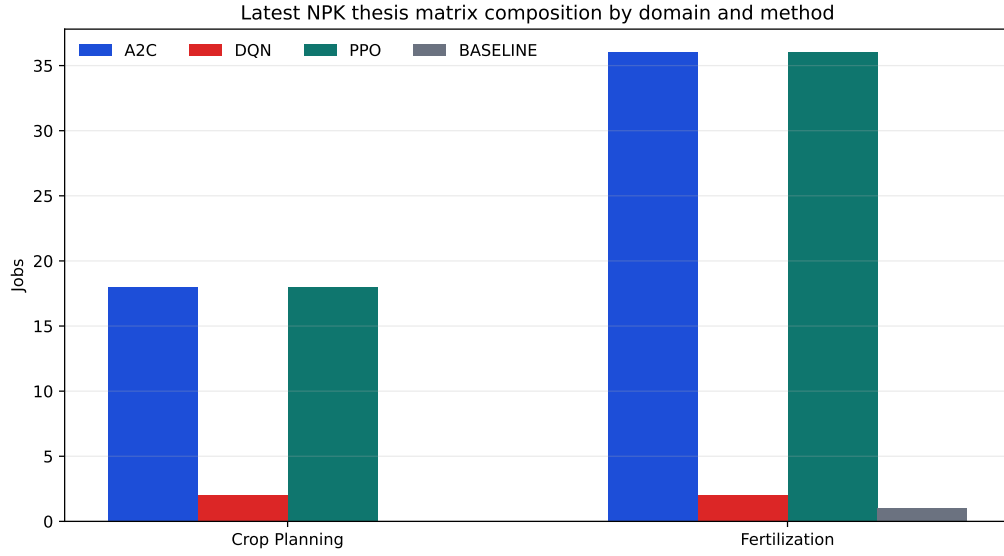


Figure 4.4: Composition of the latest broad NPK matrix by domain and algorithm family.

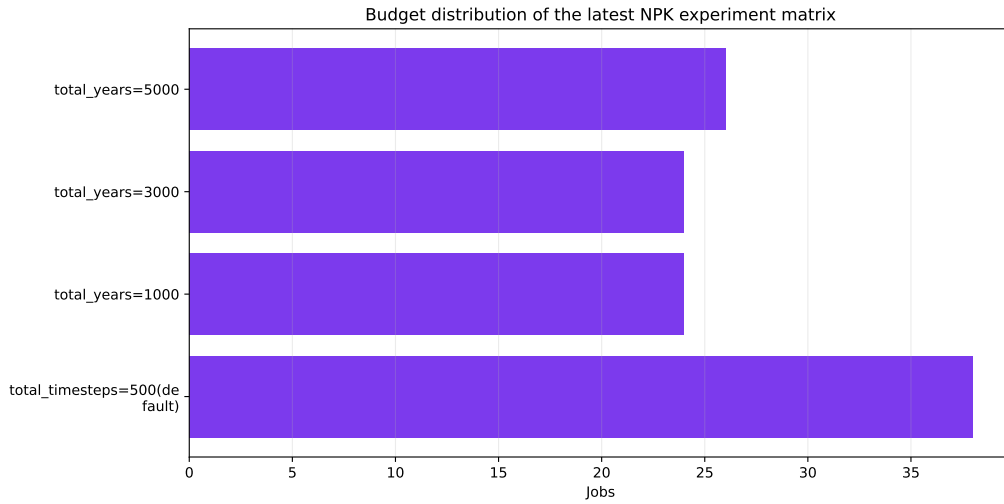


Figure 4.5: Budget distribution of the latest broad NPK matrix. Fertilization budgets are expressed in total-years, while crop-planning jobs retain the current default timestep configuration.

Table 4.3: Full broad latest-NPK matrix encoded in `run_experiments_7_3_2026.py`.

Domain	Method	Hierarchical	Jobs
crop planning	A2C	False	12
crop planning	A2C	True	6
crop planning	DQN	False	2
crop planning	PPO	False	12
crop planning	PPO	True	6
fertilization	A2C	False	36
fertilization	BASELINE	False	1
fertilization	DQN	False	2
fertilization	PPO	False	36

The matrix is large enough that it cannot be treated casually. It is the core empirical commitment of the thesis, and it is also the reason Chapter 6 remains provisional at build time. The full matrix must be executed before the final comparative claims are frozen.

The matrix also acts as a forcing function on the rest of the methodology. Because the runner produces one summary artifact per job and one aggregated CSV, each later table in the thesis can be tied back to an explicit set of matrix rows. This is a better design than ad hoc experiment accumulation because it keeps the final narrative synchronized with one authoritative campaign definition.

4.8 Hypotheses

The thesis uses the following working hypotheses.

1. PPO will remain the strongest general-purpose method for the current stack, particularly under localized NPK settings.
2. Random-weather settings will be harder to optimize but will provide more defensible generalization evidence than fixed-weather settings.
3. Hierarchical runs will improve interpretability and may improve performance where yearly crop choices materially condition weekly nutrient allocation.
4. Cost decomposition and window-compliance reporting will make the final results chapter more defensible even when aggregate returns are similar across multiple settings.

These are not treated as proven statements in the current draft. They are hypotheses encoded into the final experiment plan.

4.9 Metrics

The thesis relies on multiple metric families rather than a single score. The primary family includes deterministic and stochastic returns. The second family includes holdout-style evaluation for fertilization, where available. The third family includes reporting-derived metrics such as nutrient-wise cost totals, yearly crop decisions, and season-window compliance. The fourth family includes runner-level operational metrics such as runtime and failure signatures.

When the fresh campaign completes, these metrics should be interpreted in layers. Aggregate return answers whether one method outperformed another under the encoded reward. Nutrient-cost totals answer *how* the method achieved that performance. Yearly crop decisions answer whether strategic choices changed across algorithms or weather regimes.

Compliance metrics answer whether improved reward came at the expense of violating encoded agronomic timing guidance. This layered interpretation is necessary for a thesis defense because no single metric adequately captures the behavior of the implemented system.

This multi-metric design is consistent with the thesis goal of writing a results chapter that is analytically useful rather than merely leaderboard-oriented. A policy that attains a strong scalar reward but violates seasonal windows or relies on implausible cost behavior would not be persuasive in a thesis defense.

4.10 Claims Policy and Statistical Discipline

The thesis enforces a claims policy that is stricter than the raw availability of scores. The following claims are allowed once the broad fresh campaign completes: comparative performance between algorithms, weather modes, adaptive vs. nonadaptive settings, and hierarchical vs. non-hierarchical variants under the current Pakistan NPK stack. The following claims remain disallowed without additional work: irrigation optimization, rice-wheat empirical conclusions, field validation, and any statement that implies a global optimum over all possible agronomic policies.

This policy is a methodological choice, not an afterthought. It ensures that the thesis remains coherent across the abstract, methods, results, and limitations chapters.

Chapter 5

Implementation, Reproducibility, and Evaluation Setup

5.1 Software and Repository Stack

The current thesis stack is implemented inside a single repository that contains the CYCLES simulator interface, CyclesGym environments, experiments, demo applications, documentation, and the new thesis workspace. This co-location is useful because it keeps code, generated summaries, and writing artifacts in one versioned project. The active training flows are implemented in `experiments/fertilization/train.py` and `experiments/crop_planning/train.py`, while the unified experiment matrix is orchestrated through `run_experiments_7_3_2026.py`.

The repo has also been hardened with tests and bug fixes before the fresh campaign. This is important because reproducibility is not only about saving a seed. It is also about ensuring that the environment logic, evaluator code paths, and reporting callbacks behave consistently.

5.2 Data Inputs and Provenance

The thesis uses three categories of inputs. The first is environmental input: the Pakistan weather file and Pakistan soil file wired into the current training and evaluation defaults. The second is economic input: yearly crop and nutrient price series generated from official-source-derived pipelines. The third is operational input: runner configurations, experiment budgets, and reporting destinations.

The localized price series is especially important because it converts the cost-aware framing from a narrative claim into an executable one. Reward terms that depend on nutrient cost are no longer interpreted through a generic legacy profile alone; they can now be linked to yearly Pakistan-oriented series assembled from FAOSTAT and NFDC-linked sources [1, 10].

Table 5.1: Core artifacts that make the thesis stack reproducible.

Artifact	Path family	Role in the thesis
Pakistan weather file	<code>cycles/input/Pakistan_Site_final.weather</code>	Historical daily weather source used in training and evaluation.
Pakistan soil file	<code>cycles/input/Pakistan_Soil_final.soil</code>	Local soil profile used by the active environments.
Pakistan yearly price series	<code>cyclesgym/resources/pricing/pakistan_yearly_series.json</code>	Official-source-derived crop and nutrient price series.
Runner summary CSV	<code>runs/experiment_summaries/run_experiments_7_3_2026_summary.csv</code>	Authoritative latest-NPK experiment matrix definition.
Historical audit artifacts	<code>ExperimentationandResults/artifacts/</code>	Historical benchmark context and failure audit.
Change logs	<code>Changes/THESIS_IMPLEMENTATION/*.md</code>	Engineering traceability for thesis claims about implemented extensions.

Data provenance flow from official sources to thesis evidence

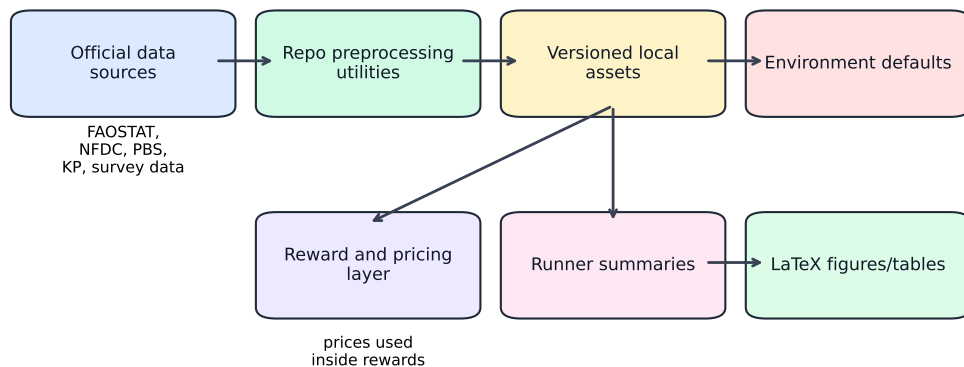


Figure 5.1: Data provenance flow from official sources to local thesis artifacts. The figure clarifies how source documents become executable weather, calendar, and pricing inputs.

Table 5.2: Data provenance summary for the localized thesis stack.

Asset family	Repo artifact	Upstream source family	Thesis role
Weather history	Pakistan_Site_final.weather	Pakistan-local weather processing pipeline informed by NASA POWER style provenance	Drives train and evaluation episodes.
Soil profile	Pakistan_Soil_final.soil	Localized soil setup with SoilGrids-style documentation lineage	Anchors the simulated site conditions.
Crop prices	pakistan_yearly_series.json crop section	FAOSTAT producer prices and local conversions	Turns yield into localized revenue.
Nutrient prices	pakistan_yearly_series.json nutrient section	NFDC fertilizer price tables and nutrient-content conversion logic	Turns N, P, and K actions into explicit cost signals.
Crop calendars	pakistan_crop_calendar.py	PBS kharif/rabi calendars and KP maize guidance	Constrains planting windows to Pakistan-relevant seasonal ranges.

5.3 Training and Evaluation Pipeline

The training pipeline used by the thesis stack is summarized in Figure 5.2. The command-line configuration determines which environment is built, how the vectorized wrapper is applied, which algorithm is trained, and where evaluation and logging artifacts are written.

Training and evaluation pipeline used for thesis experiments

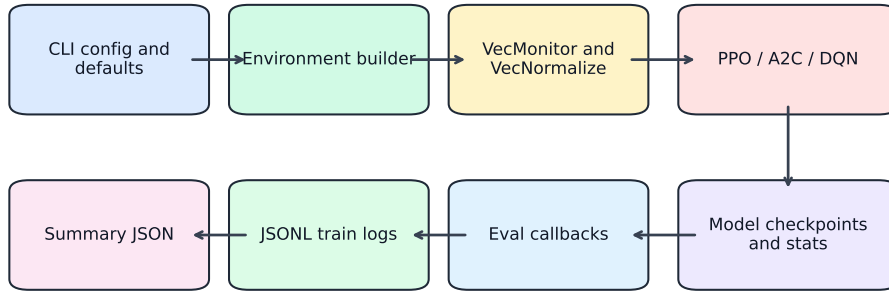


Figure 5.2: Training and evaluation pipeline used by the thesis experiments. The same pipeline is also responsible for the summary JSON outputs consumed by the writing workflow.

The use of standardized per-run summary JSON files is a major improvement over ad hoc logging because it provides a single machine-readable record of the quantities that the thesis actually cares about. This includes deterministic and stochastic returns, uplift against baseline where available, holdout-style returns for fertilization, and run metadata required to rebuild tables.

Table 5.3: Machine-readable outputs that connect experiments to thesis tables and figures.

Artifact	Producer	Main fields used by the thesis
Per-run summary JSON	training en-trypoints and unified runner	deterministic return, stochastic return mean/std, baseline uplift, holdout return, status, runtime metadata
Unified summary CSV	run_experim ents_7_3_2026.py	matrix coverage, method labels, domain, weather regime, seed, budget, status
Weekly NPK log	Hierarchical ThesisReport Callback	timestep, date, nutrient masses, nutrient costs, crop label, planner-applied flag
Yearly crop deci-sions	Hierarchical ThesisReport Callback	crop name, operation year, planting DOYs, compliance flags
Season-window compliance CSV	Hierarchical ThesisReport Callback	yearly compliance counts and aggregate compliance rate

5.4 Reporting Pipeline

The reporting pipeline is equally important because the thesis does not rely only on aggregate returns. Hierarchical environments can now write weekly NPK logs, yearly crop decisions, season-window compliance outputs, and summary JSON files. These artifacts are the bridge between the software system and the writing process.

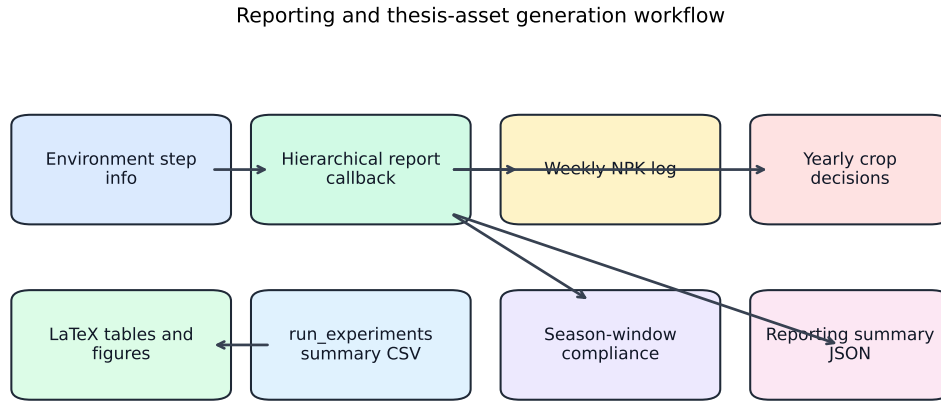


Figure 5.3: Reporting and thesis-asset generation workflow. This pipeline makes the final document reproducible from experiment outputs rather than manual transcription.

The present LaTeX workspace takes that principle seriously. Figures and tables are not manually painted in office software. They are generated from repository artifacts and written into versioned folders under `figures/generated` and `tables/generated`. This is a deliberate design choice because it reduces transcription error and makes the thesis easier to update when the fresh latest-NPK matrix completes.

5.4.1 Smoke-Run Design

Before the full latest-NPK campaign is executed, the repository includes a smoke-run stage whose purpose is operational rather than comparative. The smoke stage verifies that the training entrypoints launch correctly, summary JSONs are written, reporting callbacks emit their expected files, and the thesis asset pipeline can consume those files without manual intervention. This is the correct place to catch environment-construction bugs, missing paths, and reporting-schema regressions before committing to the 113-job campaign.

5.5 Historical Runtime Budget and Campaign Planning

The repository already contains a historical audit of earlier experiments, and these records are useful for estimating the expected cost of the fresh campaign. Figure 5.4 and Table 5.4 summarize the historical runtime profile that underpins the estimate of roughly 70–80 sequential wall-clock hours for the full broad matrix.

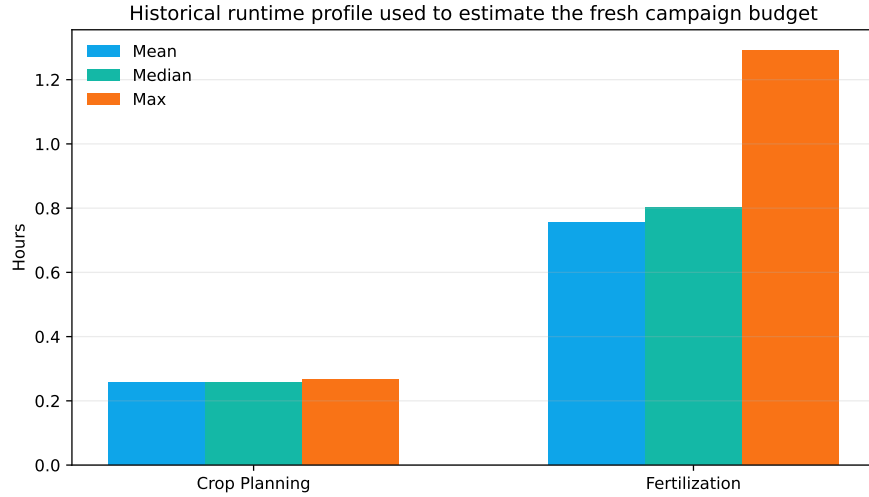


Figure 5.4: Historical runtime profile derived from earlier audited runs. This figure is used for campaign planning rather than for the main results claim.

Table 5.4: Historical runtime statistics used to estimate the wall-clock budget of the fresh broad campaign.

Domain	Mean (h)	Median (h)	Max (h)
Crop Planning	0.26	0.26	0.27
Fertilization	0.76	0.80	1.29

The thesis deliberately separates runtime planning from empirical interpretation. Historical runtime is used to plan the campaign and schedule the work. It is not treated as performance evidence for the new NPK stack.

5.6 Reproducibility Controls

The repository includes several reproducibility controls that are important for thesis quality.

- Standardized runner summaries write one JSON file per run and one aggregated CSV for the matrix.
- Offline or tracking-free execution is supported through the `-without-tracking` option.

- The codebase has already been regression-tested after the latest engineering phases.
- The thesis workspace rebuilds figures and tables directly from source artifacts.
- Administrative thesis metadata has been isolated in one file so that changes to names or forms do not propagate unpredictably through the document.

These controls do not guarantee perfect scientific reproducibility, but they significantly improve engineering reproducibility and reduce the risk of thesis-writing drift.

Table 5.5: Verification commands executed during the thesis workspace implementation on 8 March 2026.

Command	Result	What it validates
<code>python-munittestcycl esgym.tests.test_thesis_reporting-v</code>	Passed	Structured reporting files are written with the expected weekly, yearly, and compliance outputs.
<code>python-munittestcycl esgym.tests.test_pricing_utils-v</code>	Passed	Pakistan price profiles load correctly and preserve year-varying crop and NPK series.
<code>python-munittestcycl esgym.tests.test_crop_planning-v</code>	Passed	Crop-planning transitions remain stable after the thesis-localization changes.
<code>python-munittestcycl esgym.tests.test_hierarchical_env-v</code>	Passed	The hierarchical environment applies yearly planning only when intended and preserves step semantics.
<code>powershell-Execution PolicyBypass-File.\textbackslash{}ThesisMainWorking\textbackslash{}build.ps1</code>	Passed	Figures, tables, bibliography, and the full thesis PDF rebuild from versioned assets.

Table 5.5 is intentionally concrete. It records the validation commands that were actually executed during the construction of the thesis workspace. The goal is not to claim full formal verification. The goal is to demonstrate that the most thesis-critical pathways, namely pricing, reporting, crop-planning semantics, hierarchical semantics, and document rebuildability, were all checked directly.

5.7 Current Build-Time Status

At the time this draft was built, the latest broad NPK matrix had been fully materialized in runner form and dry-run summary form, but not executed end-to-end. This status is shown in Figure 5.5 and Table 6.1. The thesis therefore uses a staged writing strategy: Chapters 1–5 and Chapter 7 are built around implemented scope and reproducibility, while Chapter 6 is explicitly framed as provisional until the campaign is completed.

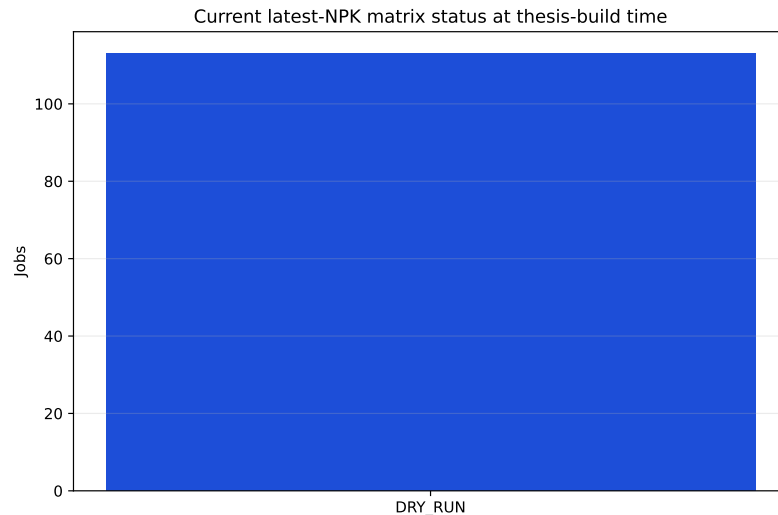


Figure 5.5: Current status of the latest broad NPK matrix at the time of this build.

Table 5.6: Current status of the latest broad NPK matrix at the time of thesis build.

Metric	Value	Interpretation
Total jobs encoded	113	The runner defines the full broad experiment campaign requested for the thesis.
Current dry-run jobs	113	The matrix has been materialized and audited, but not yet executed end-to-end at build time.
Completed latest-NPK jobs	0	No fresh broad-matrix result should be claimed as final evidence before execution.
Historical completed jobs	44	Older February 2026 runs remain useful only as historical benchmark context.

The current implementation status should therefore be read as *run-ready but results-incomplete*. That distinction matters. It means the stack is mature enough to justify a large campaign, yet the thesis still refuses to over-claim final comparative findings before the campaign artifacts exist.

5.8 Why This Chapter Matters for the Final Thesis

A thesis with strong motivation and attractive figures can still fail on committee scrutiny if the software pipeline is not reproducible. This chapter exists to prevent that weakness. It explains how the localized data, runner, reporting outputs, and rebuildable LaTeX assets fit together into a pipeline that can eventually support a final results chapter without manual patchwork.

Chapter 6

Current Results Status and Historical Benchmark Context

6.1 A Necessary Note on the State of the Main Results Chapter

The results chapter is intentionally written in a constrained form at this stage. The thesis plan locks the main empirical evidence to the latest broad Pakistan NPK matrix defined in `run_experiments_7_3_2026.py`. That matrix is already fixed, standardized, and fully enumerated, but it has not yet been executed end-to-end at the time of the current draft build. Accordingly, this chapter does not present fabricated or placeholder final comparisons. Instead, it reports three factual elements: the status of the fresh broad matrix, historical benchmark context from earlier audited runs, and the interpretation boundaries that follow from those facts.

6.2 Status of the Fresh Broad NPK Matrix

The latest broad matrix defines 113 jobs across fertilization, crop planning, hierarchical crop planning, DQN ablations, and baseline comparison paths. As of this build, the matrix exists in dry-run form and is ready to execute with standardized summary outputs.

Table 6.1: Current status of the latest broad NPK matrix at the time of thesis build.

Metric	Value	Interpretation
Total jobs encoded	113	The runner defines the full broad experiment campaign requested for the thesis.
Current dry-run jobs	113	The matrix has been materialized and audited, but not yet executed end-to-end at build time.
Completed latest-NPK jobs	0	No fresh broad-matrix result should be claimed as final evidence before execution.
Historical completed jobs	44	Older February 2026 runs remain useful only as historical benchmark context.

The significance of Table 6.1 is methodological. It prevents the thesis from quietly

mixing old completed runs with a new unexecuted design while presenting both as one coherent empirical dataset. Instead, the chapter makes the distinction explicit.

6.3 Historical Benchmark Context from the February 2026 Audit

Although the older February 2026 audited runs are not the main evidence base for the final thesis, they still provide useful historical benchmark context. They show what the repository could already achieve before the latest NPK and hierarchical engineering stack was made the primary target of the thesis campaign.

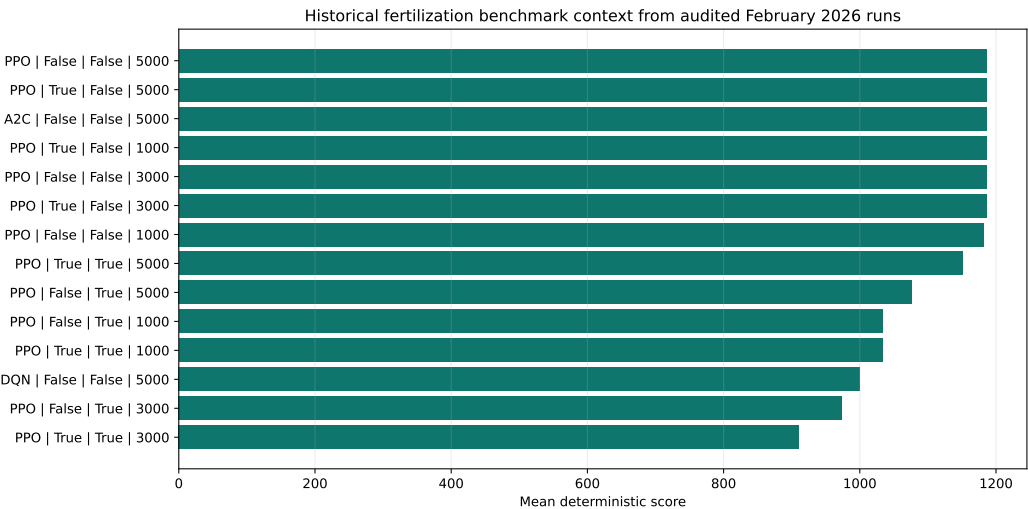


Figure 6.1: Historical fertilization benchmark context from earlier audited runs. These scores are retained only as context and should not be confused with the pending fresh broad NPK campaign.

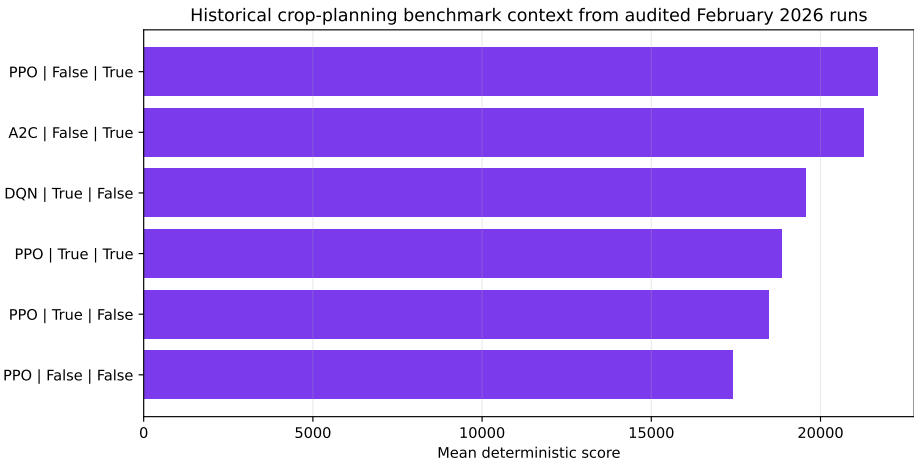


Figure 6.2: Historical crop-planning benchmark context from earlier audited runs. As above, the figure is contextual rather than final evidence for the latest stack.

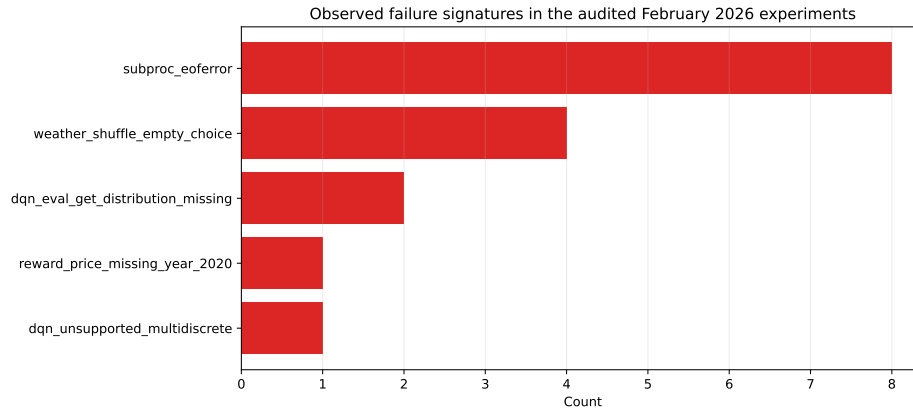


Figure 6.3: Failure signatures observed during the earlier audited experiment window. These signatures are useful for campaign planning and risk management during the fresh run phase.

Table 6.2: Historical fertilization benchmarks from audited February 2026 runs.

method	nonadaptive	fixed_weather	total_years	n_runs	mean_score	std_score	min_score	max_score
PPO	False	False	5000	3	1186.0851	0.0	1186.0851	1186.0851
PPO	True	False	5000	4	1186.0851	0.0	1186.0851	1186.0851
A2C	False	False	5000	1	1186.0851	0.0	1186.0851	1186.0851
PPO	True	False	1000	2	1186.0851	0.0	1186.0851	1186.0851
PPO	False	False	3000	2	1186.0851	0.0	1186.0851	1186.0851
PPO	True	False	3000	2	1186.0851	0.0	1186.0851	1186.0851
PPO	False	False	1000	2	1182.3651	5.2609	1178.645	1186.0851
PPO	True	True	5000	3	1150.8797	203.5457	1033.3625	1385.914
PPO	False	True	5000	3	1076.8593	290.8946	810.1625	1387.053
PPO	False	True	1000	2	1033.3625	0.0	1033.3625	1033.3625
PPO	True	True	1000	2	1033.3625	0.0	1033.3625	1033.3625
DQN	False	False	5000	1	1000.0851	0.0	1000.0851	1000.0851
PPO	False	True	3000	2	973.8426	84.1739	914.3226	1033.3625
PPO	True	True	3000	2	910.6025	173.6089	787.8425	1033.3625

Table 6.3: Historical crop-planning benchmarks from audited February 2026 runs.

method	nonadaptive	fixed_weather	n_runs	mean_score	std_score	min_score	max_score
PPO	False	True	1	21683.79	0.0	21683.79	21683.79
A2C	False	True	1	21292.441	0.0	21292.441	21292.441
DQN	True	False	1	19553.072	0.0	19553.072	19553.072
PPO	True	True	1	18848.514	0.0	18848.514	18848.514
PPO	True	False	1	18469.4	0.0	18469.4	18469.4
PPO	False	False	1	17403.996	0.0	17403.996	17403.996

Table 6.4: Historical failure signatures observed in the audited February 2026 runs.

failure_signature	count
subproc_eoferror	8
weather_shuffle_empty_choice	4
dqn_eval_get_distribution_missing	2
reward_price_missing_year_2020	1
dqn_unsupported_multidiscrete	1

The historical results support two limited but still useful observations. First, PPO historically appeared to be the strongest general-purpose algorithm in the repository. Second, operational failure modes such as subprocess interruptions and weather-window issues were not trivial and had to be addressed before treating a broad matrix as reliable. These observations helped shape the engineering priorities documented in earlier chapters.

6.4 Interpretation of Historical Trends

The historical benchmark context suggests that the repository was already capable of learning meaningful policies before the latest broad NPK campaign was defined. However, those historical runs were not produced under the full current thesis stack. They predate the final Pakistan-default NPK run-readiness phase and therefore cannot serve as the final evidence for claims about the new matrix.

This distinction is particularly important because the latest thesis framing emphasizes localized economics, NPK-aware fertilization, and hierarchical reporting. A return obtained before those features were locked in may still be informative, but it cannot be presented as if it emerged from the final target system.

6.5 What the Fresh Campaign Must Still Deliver

For the final version of this chapter, the fresh broad campaign must deliver at least four categories of output.

1. Completed per-run summary JSON files for all or clearly documented subsets of the 113 planned jobs.

2. Comparative tables for PPO, A2C, DQN, baseline, and hierarchical variants under the latest Pakistan NPK defaults.
3. Interpretable reporting outputs, including NPK cost decomposition and yearly crop-decision traces.
4. A final claims section that is statistically and narratively consistent with the completed run coverage.

Without those outputs, the thesis can describe the system and its readiness, but it cannot legitimately present the full comparative story envisioned in the plan.

6.6 Threats to Validity at Build Time

There are three main threats to validity at the current draft stage. The first is scope drift: it would be easy to accidentally write the thesis as if irrigation or rice-wheat results were already implemented. The second is evidence drift: it would be tempting to reuse older benchmark results as if they were the latest broad NPK evidence. The third is reporting drift: it would be possible to describe the system more cleanly in prose than it actually behaves in code.

The thesis addresses these threats by keeping the scope boundaries explicit, by isolating historical benchmark context from future main evidence, and by generating tables and figures from repository artifacts rather than manual restatement.

6.7 Conclusion of the Provisional Results Chapter

The present chapter should therefore be read as a disciplined staging chapter rather than as a finished empirical endpoint. It documents where the thesis currently stands, what historical evidence exists, and what still needs to happen before the final comparative narrative can be written without over-claiming. In that sense, the chapter is part of the thesis methodology as much as it is part of the results section.

Chapter 7

Discussion, Limitations, Synopsis Gap Closure, and Future Work

7.1 What the Thesis Has Actually Achieved

The thesis has achieved a substantial transformation of the original CyclesGym base repository. It now supports Pakistan-adapted weather and soil inputs, Pakistan-oriented yearly price series, NPK-aware fertilization logic, hierarchical crop-planning integration, and structured reporting suitable for thesis writing. These are not superficial modifications. They change the scope, interpretability, and scientific defensibility of the system.

A core strength of the work is that it treats engineering as part of the research contribution. In simulator-based RL projects, progress is often described mainly through final returns. That is insufficient here. The thesis demonstrates that a credible agricultural RL platform also requires localized data provenance, explicit scope control, standardized outputs, and enough testing discipline that the final experiment campaign can be trusted when it runs.

Another achievement is conceptual clarification. The current repository now draws a sharper line between the role of the simulator, the role of RL, and the role of academic reporting. The simulator provides the transition dynamics. RL provides a way to optimize sequential decisions under those dynamics. The reporting layer provides the evidence needed to explain what the optimizer actually did. Treating these three roles separately has improved both the codebase and the thesis narrative.

7.2 Why Scope Control is a Scientific Strength

The most visible limitation of the thesis is also one of its strengths: it refuses to claim completion of everything promised in the synopsis. Irrigation as a learned action is not implemented. Rice-specific localization is not implemented. The main experimental crop setup remains maize-soy. The full broad latest-NPK matrix has been defined but not yet completed at build time. These facts narrow the current empirical chapter, but they also protect the thesis from making indefensible claims.

In many research settings, the temptation is to smooth over these gaps by writing a broader story than the code can support. This thesis takes the opposite approach. It uses a synopsis closure matrix to make the gaps visible and to distinguish clearly between completed engineering, partial implementation, and future work.

Table 7.1: Synopsis closure matrix used to keep thesis claims aligned with the implemented repo state.

Synopsis item	Status	Evidence in the current repo	Thesis treatment
Pakistan-adapted weather and soil inputs	Implemented	Pakistan weather and soil files are wired into train and evaluation entrypoints.	Present as a completed contribution.
Cost-aware reward shaping	Implemented	Reward stack and Pakistan price profiles are implemented and tested.	Present as a completed contribution.
Multi-nutrient fertilizer modeling	Implemented	NPK action mode, price series, and reporting are implemented.	Present as a completed contribution.
Hierarchical RL across planning and fertilization	Implemented	Dedicated hierarchical environment and reporting callback exist.	Present as a completed contribution with explicit scope bounds.
Irrigation as a learned action	Not implemented	No active training flow exposes irrigation as a learned control variable.	Present as future work only.
Rice-specific cultivar localization	Not implemented	Current crop-planning stack uses the working maize-soy configuration.	Present as future work only.
Rice-wheat experimental rotation	Not implemented	Crop-calendar support exists, but the main experiment setup remains maize-soy.	Present as a deferred extension, not a result claim.
Farmer-facing advisory rule export	Partial	CSV-style reporting exists, but no polished advisory layer is implemented.	Present as partial engineering output.
Formal price- and climate-shock campaign	Partial	The matrix runner is ready, but the broad fresh NPK campaign is still pending at build time.	Present as a planned experiment campaign.

Scope control also has a practical defense benefit. It allows difficult questions to be answered directly. If a committee asks whether the thesis optimizes irrigation, the answer is no. If it asks whether the thesis completed a Pakistan-wide rice-wheat evaluation, the answer is no. If it asks whether the thesis nevertheless built a more localized, cost-aware,

and more interpretable CyclesGym-derived platform, the answer is yes. This asymmetry is acceptable because a thesis is judged on defensible contribution, not on maximal ambition alone.

7.3 Limitations

The thesis has several technical and scientific limitations.

- The main broad latest-NPK campaign remains pending, so Chapter 6 is currently provisional.
- The active crop-planning and hierarchical experiments are still tied to the current working maize-soy configuration rather than a rice-wheat configuration.
- Irrigation has not yet been exposed as a learned control variable in the active training flows.
- The results remain simulation-based and have not yet been field-validated.
- Economic realism has improved substantially, but broader farm-level costs and market frictions are still simplified.

These limitations do not invalidate the completed work, but they define the boundary of what the thesis can responsibly claim.

Table 7.2: Limitations matrix used to distinguish current evidence boundaries from future work.

Limitation	Category	Why it matters	Handling in thesis
Fresh 113-job campaign not yet completed	Evidence	Final algorithm comparisons cannot yet be frozen	Keep Chapter 6 provisional and clearly staged.
Working crop setup remains maize-soy	Scope	Limits generalization to broader Pakistan rotations	Present as current experimental configuration, not universal recommendation.
Irrigation not exposed as learned action	Control design	Proposal-scale water-management claims are unsupported	Treat as future work only.
No field validation	External validity	Simulator success does not guarantee farm deployment success	State explicitly in limitations and roadmap.
Simplified economics beyond prices	Modeling realism	Profit-related reward omits wider farm logistics and market frictions	Limit claims to cost-aware simulation, not full farm economics.

The limitations matrix is included because not all limitations have the same character. Some are evidence limitations, such as a pending experiment campaign. Others are modeling limitations, such as simplified economics. Others are scope limitations inherited from the fact that the thesis chose to stabilize one working crop configuration instead of spreading effort across many unstable configurations. Treating all limitations as one undifferentiated list would make later prioritization weaker.

7.4 Implications for a Thesis Defense

From a defense perspective, the thesis is strongest when it is presented as a system-construction and experimental-readiness contribution grounded in agricultural RL. It can credibly claim that the original base repository has been transformed into a localized and

more interpretable research platform. It can also credibly explain why the fresh campaign must be executed before final comparative conclusions are frozen. This combination of technical depth and scope honesty is preferable to a superficially broader but less defensible narrative.

The historical benchmark context can still be useful in a defense because it shows that the repository is not a blank slate. Yet the defense should repeatedly emphasize that older runs are contextual and that the main thesis evidence is locked to the latest broad NPK design.

A disciplined defense posture therefore has two parts. First, it demonstrates mastery of the implemented system: environments, rewarders, pricing utilities, reporting outputs, and runner semantics. Second, it avoids claiming that simulator-based performance is equivalent to field-ready agronomy. This balance is likely to be viewed as maturity rather than weakness, especially in a technically demanding interdisciplinary topic.

7.5 Future Work

Future work follows naturally from the current limitations.

1. Complete the full broad latest-NPK matrix and regenerate all Chapter 6 figures from the completed summaries.
2. Add irrigation as a learned action so that the thesis scope aligns more closely with the original synopsis.
3. Retarget the crop-planning stack from the current working maize-soy setup toward a Pakistan-focused cereal rotation if agronomic data and configuration stability permit.
4. Add stronger statistical analysis, including confidence intervals and formal significance testing across seeds where coverage is sufficient.
5. Conduct expert-in-the-loop or field-adjacent validation to test whether the learned policies remain plausible outside the simulator.

Roadmap implied by the current thesis limitations

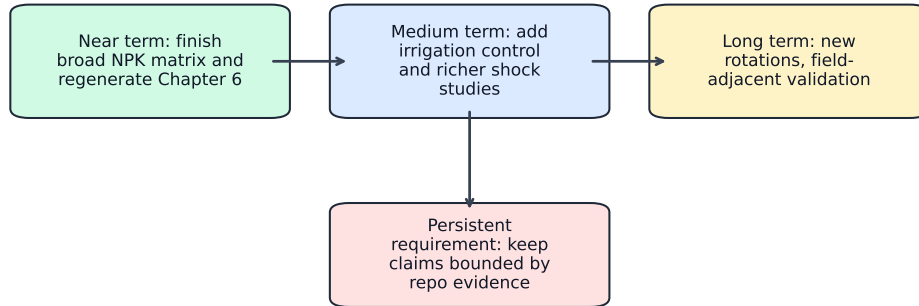


Figure 7.1: Roadmap implied by the current thesis limitations. The roadmap is staged so that empirical completion, control expansion, and external validation occur in a defensible order.

The order shown in Figure 7.1 is intentional. The thesis should not jump to new crops, irrigation, or validation studies before completing the already committed broad NPK campaign on the stabilized working stack. Otherwise, the project risks accumulating new branches of scope faster than it closes the main one.

7.6 Final Reflection

The central message of this thesis is not that agricultural RL is solved. It is that a vague idea of cost-aware crop management has been converted into a concrete, localized, testable, and reportable software stack. Even before the final broad campaign completes, that transformation has scientific value. It turns an inherited simulator wrapper into a research platform with clearer provenance, better scope control, and a more rigorous path to defensible experimentation. Once the fresh broad campaign is executed, the thesis can add the full comparative chapter on top of this foundation without needing to rewrite the underlying story.

Chapter A

Full Broad Latest-NPK Matrix Listing

This appendix lists the full broad latest-NPK matrix as currently encoded in the unified runner. The table is generated directly from the summary CSV so that the thesis plan, experiment commands, and appendix remain synchronized.

Table A.1: Full broad latest-NPK matrix listing generated from the unified runner summary CSV.

Idx	Domain	Method	Adaptive	Hierarchical	Weather	Budget
1	fertilization	PPO	True	False	True	total_years=1000
2	fertilization	PPO	False	False	True	total_years=1000
3	fertilization	PPO	True	False	False	total_years=1000
4	fertilization	PPO	False	False	False	total_years=1000
5	fertilization	PPO	True	False	True	total_years=1000
6	fertilization	PPO	False	False	True	total_years=1000
7	fertilization	PPO	True	False	False	total_years=1000
8	fertilization	PPO	False	False	False	total_years=1000
9	fertilization	PPO	True	False	True	total_years=1000
10	fertilization	PPO	False	False	True	total_years=1000
11	fertilization	PPO	True	False	False	total_years=1000
12	fertilization	PPO	False	False	False	total_years=1000
13	fertilization	PPO	True	False	True	total_years=3000
14	fertilization	PPO	False	False	True	total_years=3000
15	fertilization	PPO	True	False	False	total_years=3000
16	fertilization	PPO	False	False	False	total_years=3000
17	fertilization	PPO	True	False	True	total_years=3000
18	fertilization	PPO	False	False	True	total_years=3000
19	fertilization	PPO	True	False	False	total_years=3000
20	fertilization	PPO	False	False	False	total_years=3000
21	fertilization	PPO	True	False	True	total_years=3000
22	fertilization	PPO	False	False	True	total_years=3000
23	fertilization	PPO	True	False	False	total_years=3000

Idx	Domain	Method	Adaptive	Hierarchical	Weather	Budget
24	fertilization	PPO	False	False	False	total_years=3000
25	fertilization	PPO	True	False	True	total_years=5000
26	fertilization	PPO	False	False	True	total_years=5000
27	fertilization	PPO	True	False	False	total_years=5000
28	fertilization	PPO	False	False	False	total_years=5000
29	fertilization	PPO	True	False	True	total_years=5000
30	fertilization	PPO	False	False	True	total_years=5000
31	fertilization	PPO	True	False	False	total_years=5000
32	fertilization	PPO	False	False	False	total_years=5000
33	fertilization	PPO	True	False	True	total_years=5000
34	fertilization	PPO	False	False	True	total_years=5000
35	fertilization	PPO	True	False	False	total_years=5000
36	fertilization	PPO	False	False	False	total_years=5000
37	fertilization	A2C	True	False	True	total_years=1000
38	fertilization	A2C	False	False	True	total_years=1000
39	fertilization	A2C	True	False	False	total_years=1000
40	fertilization	A2C	False	False	False	total_years=1000
41	fertilization	A2C	True	False	True	total_years=1000
42	fertilization	A2C	False	False	True	total_years=1000
43	fertilization	A2C	True	False	False	total_years=1000
44	fertilization	A2C	False	False	False	total_years=1000
45	fertilization	A2C	True	False	True	total_years=1000
46	fertilization	A2C	False	False	True	total_years=1000
47	fertilization	A2C	True	False	False	total_years=1000
48	fertilization	A2C	False	False	False	total_years=1000
49	fertilization	A2C	True	False	True	total_years=3000
50	fertilization	A2C	False	False	True	total_years=3000
51	fertilization	A2C	True	False	False	total_years=3000
52	fertilization	A2C	False	False	False	total_years=3000
53	fertilization	A2C	True	False	True	total_years=3000
54	fertilization	A2C	False	False	True	total_years=3000
55	fertilization	A2C	True	False	False	total_years=3000
56	fertilization	A2C	False	False	False	total_years=3000
57	fertilization	A2C	True	False	True	total_years=3000
58	fertilization	A2C	False	False	True	total_years=3000
59	fertilization	A2C	True	False	False	total_years=3000
60	fertilization	A2C	False	False	False	total_years=3000

Idx	Domain	Method	Adaptive	Hierarchical	Weather	Budget
61	fertilization	A2C	True	False	True	total_years=5000
62	fertilization	A2C	False	False	True	total_years=5000
63	fertilization	A2C	True	False	False	total_years=5000
64	fertilization	A2C	False	False	False	total_years=5000
65	fertilization	A2C	True	False	True	total_years=5000
66	fertilization	A2C	False	False	True	total_years=5000
67	fertilization	A2C	True	False	False	total_years=5000
68	fertilization	A2C	False	False	False	total_years=5000
69	fertilization	A2C	True	False	True	total_years=5000
70	fertilization	A2C	False	False	True	total_years=5000
71	fertilization	A2C	True	False	False	total_years=5000
72	fertilization	A2C	False	False	False	total_years=5000
73	crop planning	PPO	True	False	True	total_timesteps=500(default)
74	crop planning	PPO	False	False	True	total_timesteps=500(default)
75	crop planning	PPO	hierarchical	True	True	total_timesteps=500(default)
76	crop planning	PPO	True	False	False	total_timesteps=500(default)
77	crop planning	PPO	False	False	False	total_timesteps=500(default)
78	crop planning	PPO	hierarchical	True	False	total_timesteps=500(default)
79	crop planning	PPO	True	False	True	total_timesteps=500(default)
80	crop planning	PPO	False	False	True	total_timesteps=500(default)
81	crop planning	PPO	hierarchical	True	True	total_timesteps=500(default)
82	crop planning	PPO	True	False	False	total_timesteps=500(default)
83	crop planning	PPO	False	False	False	total_timesteps=500(default)
84	crop planning	PPO	hierarchical	True	False	total_timesteps=500(default)
85	crop planning	PPO	True	False	True	total_timesteps=500(default)
86	crop planning	PPO	False	False	True	total_timesteps=500(default)
87	crop planning	PPO	hierarchical	True	True	total_timesteps=500(default)
88	crop planning	PPO	True	False	False	total_timesteps=500(default)
89	crop planning	PPO	False	False	False	total_timesteps=500(default)
90	crop planning	PPO	hierarchical	True	False	total_timesteps=500(default)
91	crop planning	A2C	True	False	True	total_timesteps=500(default)
92	crop planning	A2C	False	False	True	total_timesteps=500(default)
93	crop planning	A2C	hierarchical	True	True	total_timesteps=500(default)
94	crop planning	A2C	True	False	False	total_timesteps=500(default)
95	crop planning	A2C	False	False	False	total_timesteps=500(default)
96	crop planning	A2C	hierarchical	True	False	total_timesteps=500(default)
97	crop planning	A2C	True	False	True	total_timesteps=500(default)

Idx	Domain	Method	Adaptive	Hierarchical	Weather	Budget
98	crop planning	A2C	False	False	True	total_timesteps=500(default)
99	crop planning	A2C	hierarchical	True	True	total_timesteps=500(default)
100	crop planning	A2C	True	False	False	total_timesteps=500(default)
101	crop planning	A2C	False	False	False	total_timesteps=500(default)
102	crop planning	A2C	hierarchical	True	False	total_timesteps=500(default)
103	crop planning	A2C	True	False	True	total_timesteps=500(default)
104	crop planning	A2C	False	False	True	total_timesteps=500(default)
105	crop planning	A2C	hierarchical	True	True	total_timesteps=500(default)
106	crop planning	A2C	True	False	False	total_timesteps=500(default)
107	crop planning	A2C	False	False	False	total_timesteps=500(default)
108	crop planning	A2C	hierarchical	True	False	total_timesteps=500(default)
109	fertilization	DQN	True	False	True	total_years=5000
110	fertilization	DQN	True	False	False	total_years=5000
111	crop planning	DQN	False	False	True	total_timesteps=500(default)
112	crop planning	DQN	False	False	False	total_timesteps=500(default)
113	fertilization	BASELINE	nan	False	nan	nan

Chapter B

Reproducibility and Build Notes

B.1 Thesis Build Commands

The LaTeX workspace can be rebuilt by running the following command from the repository root:

```
powershell -ExecutionPolicy Bypass -File .\Thesis Main Working\build.ps1
```

This command regenerates all figures and tables from repository artifacts and then compiles the thesis using the portable Tectonic binary stored under `.tools/tectonic/`.

B.2 Smoke-Run Script

A smoke-run script has been included to launch a small set of real runs before the full broad campaign is started:

```
powershell -ExecutionPolicy Bypass -File .\Thesis Main Working\scripts\run_smoke.ps1
```

The script targets three representative cases: one fertilization PPO run, one crop-planning PPO run, and one hierarchical PPO run. The outputs are directed to `Thesis Main Working/notes/smoke_runs/`.

B.3 Why Smoke Runs Precede the Full Matrix

The smoke stage is deliberately narrower than the full latest-NPK matrix. Its role is to verify that the major execution pathways are alive: environment creation, CYCLES invocation, summary-file writing, hierarchical report generation, and thesis-asset regeneration. This prevents the main experiment campaign from becoming an expensive debugging session. In a simulator-heavy RL repo, this staging discipline is part of the reproducibility methodology rather than an optional convenience.

B.4 Why the Current Draft Keeps Chapter 6 Provisional

The latest broad NPK matrix is the intended main evidence base for the thesis. Because that campaign has not completed at the time of this build, the present draft treats Chapter 6 as a constrained status-and-context chapter rather than as the final results chapter. This is a deliberate integrity control, not a missing feature.

B.5 Administrative Metadata

University-specific metadata, including title-page fields and signatory names, has been isolated in `frontmatter/metadata.tex`. This allows rapid correction of administrative details before final submission without editing the technical chapters.

B.6 Rebuild Philosophy

The thesis workspace follows a “generated, not hand-copied” philosophy. Tables are emitted as standalone `.tex` snippets. Plots and diagrams are generated by Python scripts and versioned as PDF assets. The build script composes these generated artifacts with the chapter text. This design does not make the thesis immune to error, but it materially reduces the risk of stale tables, inconsistent labels, and manually transcribed result drift.

Chapter C

Source and Artifact Map

This appendix summarizes the main source families that ground the thesis.

- Official formatting and university sources: the official NUST MS thesis template, NUST Students Hub, and SEECS downloads pages.
- Pakistan context and economics: Pakistan Economic Survey, FAOSTAT producer prices, NFDC fertilizer prices, PBS crop calendars, and KP maize guidance.
- Technical foundations: CyclesGym, CropGym, gym-DSSAT, the agricultural RL review, and supporting RL literature.
- Local artifact sources: repository change logs, experiment audits, runner summaries, and generated pricing data files.

The explicit purpose of this appendix is to document that the thesis is grounded both in external primary sources and in local implementation artifacts. The writing process therefore remains traceable even when the repository continues to evolve.

Table C.1: File-level responsibilities of the core thesis components.

Component	Main file	Responsibility	Why it matters to the thesis
Weekly fertilization environment	<code>cyclesgym/envs/corn.py</code>	Maps weekly agent actions into fertilizer operations and simulator runs.	Main benchmark for Pakistan-aligned fertilization experiments.
Flat crop-planning environment	<code>cyclesgym/envs/crop/_planning.py</code>	Encodes annual crop and planting choices with soil-nitrogen observations.	Strategic planning layer for the maize-soy thesis setup.
Hierarchical environment	<code>cyclesgym/envs/hierarchical.py</code>	Couples yearly crop planning with weekly NPK fertilization and rich info dictionaries.	Core thesis-specific system extension.
Pricing localization	<code>cyclesgym/utils/pricing/_utils.py</code>	Loads Pakistan yearly crop and nutrient price series with legacy fallbacks.	Converts narrative cost-awareness into executable reward logic.
Reporting callback	<code>cyclesgym/utils/thesis/_reporting.py</code>	Writes weekly nutrient logs, yearly crop decisions, compliance files, and summary JSON.	Makes Chapter 6 traceable to generated artifacts.
Experiment orchestrator	<code>run/_experiments/_7/_3/_2026.py</code>	Builds the 113-job matrix and standardizes per-run summary outputs.	Defines the authoritative latest-NPK campaign.

C.1 How to Read the Source Map

External sources justify context, methodology, and localization choices. Local artifacts justify claims about implementation state, experiment readiness, and reproducibility. Neither category can replace the other. This appendix is therefore intended as a quick reference for tracing any thesis claim back to either a cited source family or a versioned repository component.

Chapter D

Verification Snapshot

This appendix records the targeted verification work carried out while constructing the thesis workspace. The purpose is not to claim comprehensive software assurance. The purpose is to document that the most thesis-critical pathways were executed directly and observed to pass.

Table D.1: Verification commands executed during the thesis workspace implementation on 8 March 2026.

Command	Result	What it validates
<code>python-munittestcycl esgym.tests.test_th esis_reporting-v</code>	Passed	Structured reporting files are written with the expected weekly, yearly, and compliance outputs.
<code>python-munittestcycl esgym.tests.test_pr icing_utils-v</code>	Passed	Pakistan price profiles load correctly and preserve year-varying crop and NPK series.
<code>python-munittestcycl esgym.tests.test_cr op_planning-v</code>	Passed	Crop-planning transitions remain stable after the thesis-localization changes.
<code>python-munittestcycl esgym.tests.test_hi erarchical_env-v</code>	Passed	The hierarchical environment applies yearly planning only when intended and preserves step semantics.
<code>powershell-Execution PolicyBypass-File.\t extbackslash{}Thesi sMainWorking\textbac kslash{}build.ps1</code>	Passed	Figures, tables, bibliography, and the full thesis PDF rebuild from versioned assets.

D.1 Interpretation of the Verification Checks

The verification commands cover three different risk classes. Pricing and reporting tests protect the thesis against silent errors in the localized reward and artifact-writing layers. Crop-planning and hierarchical tests protect the semantics of the environments that the thesis discusses in Chapters 3 and 4. The document rebuild check protects the writing

pipeline itself by confirming that the thesis PDF can be regenerated from versioned assets and bibliography files.

D.2 Residual Verification Gaps

These checks do not replace the full fresh experiment campaign, and they do not prove agronomic validity. They are engineering checks on the correctness of code pathways and artifact generation. The main empirical risk remains the completion and audit of the latest broad NPK matrix, which is why the results chapter is still staged as provisional.

Chapter E

Data Provenance and Output Schema

This appendix consolidates the external-to-local data flow used by the thesis stack and the machine-readable outputs that later chapters depend on.

Table E.1: Data provenance summary for the localized thesis stack.

Asset family	Repo artifact	Upstream source family	Thesis role
Weather history	Pakistan_Site _final.weather	Pakistan-local weather processing pipeline informed by NASA POWER style provenance	Drives train and evaluation episodes.
Soil profile	Pakistan_Soil _final.soil	Localized soil setup with SoilGrids-style documentation lineage	Anchors the simulated site conditions.
Crop prices	pakistan_yearly_series.json crop section	FAOSTAT producer prices and local conversions	Turns yield into localized revenue.
Nutrient prices	pakistan_yearly_series.json nutrient section	NFDC fertilizer price tables and nutrient-content conversion logic	Turns N, P, and K actions into explicit cost signals.
Crop calendars	pakistan_crop_calendar.py	PBS kharif/rabi calendars and KP maize guidance	Constrains planting windows to Pakistan-relevant seasonal ranges.

Table E.2: Machine-readable outputs that connect experiments to thesis tables and figures.

Artifact	Producer	Main fields used by the thesis
Per-run summary JSON	training en-trypoints and unified runner	deterministic return, stochastic return mean/std, baseline uplift, holdout return, status, runtime metadata
Unified summary CSV	run_experim ents_7_3_2026.py	matrix coverage, method labels, domain, weather regime, seed, budget, status
Weekly NPK log	Hierarchical ThesisReport Callback	timestep, date, nutrient masses, nutrient costs, crop label, planner-applied flag
Yearly crop decisions	Hierarchical ThesisReport Callback	crop name, operation year, planting DOYs, compliance flags
Season-window compliance CSV	Hierarchical ThesisReport Callback	yearly compliance counts and aggregate compliance rate

E.1 Why Provenance and Schema Belong in the Thesis

In this project, data provenance and output schema are not peripheral implementation details. They are part of the scientific contract of the thesis. A localized price-aware reward is only defensible if the reader can understand where the price series came from. A results table is only reproducible if the reader can understand which JSON or CSV fields it was derived from. Including this appendix makes those dependencies explicit and reduces the distance between the written document and the actual repository artifacts.

Bibliography

- [1] Food and Agriculture Organization of the United Nations. FAOSTAT Prices: Producer Prices Bulk Download, 2026. URL [https://fenixservices.fao.org/faostat/static/bulkdownloads/Prices_E_All_Data_\(Normalized\).zip](https://fenixservices.fao.org/faostat/static/bulkdownloads/Prices_E_All_Data_(Normalized).zip). Accessed: 2026-03-08.
- [2] R. Gautron, Odalric-Ambrym Maillard, P. Preux, M. Corbeels, and R. Sabbadin. Reinforcement learning for crop management support: Review, prospects and challenges. *Computers and Electronics in Agriculture*, 200:107182, 2022. doi: 10.1016/j.compag.2022.107182. URL <https://www.sciencedirect.com/science/article/pii/S0168169922003790>.
- [3] R. Gautron, E. Padron, P. Preux, J. Bigot, Odalric-Ambrym Maillard, and D. Emukpere. gym-dssat: A crop model turned into a reinforcement learning environment. Technical Report RR-9460, Inria, 2022. URL <https://hal.inria.fr/hal-03719898>. Accessed: 2026-03-08.
- [4] Government of Khyber Pakhtunkhwa, Agriculture Department. Maize production technology, 2026. URL <https://zarat.kp.gov.pk/assets/uploads/publications/Maize%20Production%20Technology.pdf>. Accessed: 2026-03-08.
- [5] Government of Pakistan, Finance Division. Pakistan economic survey 2024-25: Agriculture chapter, 2025. URL https://www.finance.gov.pk/survey/chapter_25/02_Agriculture.pdf. Accessed: 2026-03-08.
- [6] S. Huang, A. Abdolmaleki, G. Vezzani, P. Brakel, D. J. Mankowitz, M. Neunert, S. Bohez, Y. Tassa, N. Heess, and M. Riedmiller. A constrained multi-objective reinforcement learning framework. In *Conference on Robot Learning*, pages 883–893, 2022. URL <https://proceedings.mlr.press/v205/huang23a.html>.
- [7] ISRIC World Soil Information. Soilgrids documentation, 2026. URL <https://docs.isric.org/globaldata/soilgrids/>. Accessed: 2026-03-08.
- [8] M. G. Kallenberg, H. Overweg, R. van Bree, and Ioannis N. Athanasiadis. Nitrogen management with reinforcement learning and crop growth models. *Environmental Data Science*, 2:e34, 2023. doi: 10.1017/eds.2023.30. URL <https://www.cambri>

dge.org/core/journals/environmental-data-science/article/nitrogen-management-with-reinforcement-learning-and-crop-growth-models/358749FAFAA4990B1448DAB7F48D641C.

- [9] National Aeronautics and Space Administration. NASA POWER documentation, 2026. URL <https://power.larc.nasa.gov/docs/>. Accessed: 2026-03-08.
- [10] National Fertilizer Development Centre, Pakistan. Historical fertilizer prices, 2026. URL <https://nfdc.gov.pk/Web-Page%20Updating/prices.htm>. Accessed: 2026-03-08.
- [11] Pakistan Bureau of Statistics. Approved crop calendar (Kharif), 2026. URL https://www.pbs.gov.pk/sites/default/files//tables/table_13_approved_crop_calendar_kharif.pdf. Accessed: 2026-03-08.
- [12] Pakistan Bureau of Statistics. Approved crop calendar (Rabi), 2026. URL https://www.pbs.gov.pk/sites/default/files/tables/table_14_approved_crop_calendar_rabi.pdf. Accessed: 2026-03-08.
- [13] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018. URL <http://incompleteideas.net/book/the-book-2nd.html>.
- [14] Matteo Turchetta, Luca Corinzia, Scott Sussex, Amanda Burton, Juan Herrera, Ioannis N. Athanasiadis, Joachim M. Buhmann, and Andreas Krause. Learning long-term crop management strategies with CyclesGym. In *NeurIPS 2022 Datasets and Benchmarks Track*, 2022. URL <https://openreview.net/forum?id=Zx5qJzNesn0>. Accessed: 2026-03-08.
- [15] W. Zhao, T. He, R. Chen, T. Wei, and C. Liu. State-wise safe reinforcement learning: A survey. *arXiv preprint arXiv:2302.03122*, 2023. URL <https://arxiv.org/abs/2302.03122>.